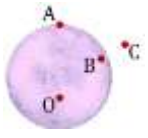
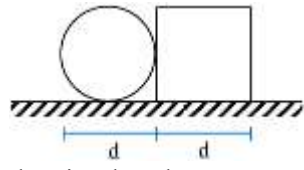


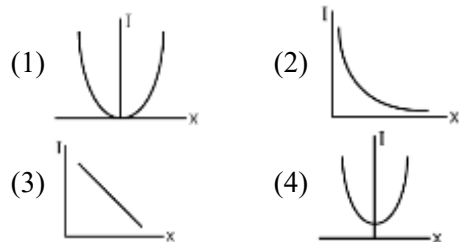
PHYSICS

1. Suppose, the gravitational attraction of the earth, suddenly disappears which of the following statement will be true
 - (1) Weight of the body will become zero but the mass will remain unchanged
 - (2) Weight of the body will remain unchanged but the mass will become zero
 - (3) Both mass as well as weight will be zero
 - (4) Neither mass nor weight will be zero
2. The length of simple pendulum for a given time period is ℓ if the pendulum is taken to a place where acceleration due to gravity is doubled, then for the time period to remain the same its length should be:
 - (1) 2ℓ (2) ℓ (3) $\ell/2$ (4) $\frac{1}{4}\ell$
3. One goes from the centre of the earth to an altitude half the radius of the earth, where will the g be the greatest?
 - (1) Center of the earth
 - (2) At a depth half the radius of the earth
 - (3) At the surface of the earth
 - (4) At an altitude equal to half the radius of the earth
4. Given that the acceleration due to gravity at a height h is same as that at the depth d below the surface of the earth. If both h and d are small as compared to the radius of the earth. Then which of the following relations is correct ?
 - (1) $2h = d$ (2) $h = d$
 - (3) $h = 2d$ (4) $h = d^2$
5. If the radius of the earth decreases by 10% the mass remaining unchanged, what will happen to the acceleration due to gravity ?
 - (1) Decreases by 19%
 - (2) Increases by 19%
 - (3) Decreases by more than 19%
 - (4) increases by more than 19%
6. At what angle with the horizontal should a projectile be fire from the top of Mount Everest, with the escape velocity to enable it escape from gravitational pull of the earth ?
 - (1) Less than 45° (2) 45°
 - (3) More than 45° (4) any angle
7. The escape velocity for a body projected vertically upwards from the surface of earth is 11 km/s. If the body is projected at 45° with the vertical, the escape velocity will be
 - (1) $11/\sqrt{2}$ km/s (2) $11/\sqrt{3}$ km/s
 - (3) 22 km/s (4) 11 km/s
8. The acceleration due to gravity on the planet A is 9 times the acceleration due to gravity on planet B. A man jumps to a height of 2m on the surface of A what is the height of jump by the same person on the planet B :
 - (1) 18m (2) 6m (3) $2/3$ m (4) $2/9$ m
9. In the given figure there is a hollow sphere then on which points gravitational potential due to the sphere will be same:
 
 - (1) O, A, B (2) O, A, B, C
 - (3) O, B (4) No where
11. The moment of inertia of a circular ring about an axis passing through its centre and normal to its plane is 200 g cm^2 then its moment of inertia about a diameter is
 - (1) 400 g cm^2 (2) 300 g cm^2
 - (3) 200 g cm^2 (4) 100 g cm^2
11. A constant torque acting on a uniform circular wheel changes its angular momentum from A_0 to $4 A_0$ in 4 seconds. The magnitude of this torque is
 - (1) $3A_0/4$ (2) A_0
 - (3) $4A_0$ (4) $12 A_0$
12. The ratio of diameter of two satellites is p and the ratio of their acceleration due to gravity is q . The ratio of their escape velocities will be :
 - (1) $\left(\frac{q}{p}\right)^{\frac{1}{2}}$ (2) $\left(\frac{p}{q}\right)^{\frac{1}{2}}$ (3) pq (4) $\sqrt{pq}$
13. A ring of mass m and radius r rotates about an axis passing through its centre and perpendicular to its plane with angular velocity ω . Its kinetic energy is
 - (1) $\frac{1}{2} mr^2 \omega^2$ (2) $mr\omega^2$
 - (3) $mr^2 \omega^2$ (4) $\frac{1}{2} mr\omega^2$
14. A circular plate of diameter d is kept in contact with a square plate of edge d as shown in figure. The density of the material and the thickness are same everywhere. The centre of mass of the composite system will be:
 
 - (1) inside the circular plate
 - (2) outside the system
 - (3) at the point of contact
 - (4) inside the square plate

15. Three point masses each of mass m are placed at the corners of an equilateral triangle of side b . The moment of inertia of the system about an axis coinciding with one side of triangle is :

(1) $3mb^2$ (2) mb^2
(3) $(3/4)mb^2$ (4) $(2/3)mb^2$

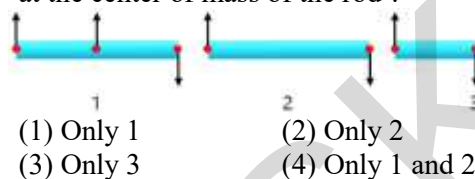
16. The curve for the moment of inertia of a disc of constant mass M versus distance of axis from its centre?



17. A ball of mass 0.25 kg attached to the end of a string of length 1.96 m is moving in a horizontal circle. The string will break if the tension is more than 25 N . What is the maximum speed with which the ball can be moved?

(1) 14 m/s (2) 3 m/s
(3) 392 m/s (4) 5 m/s

18. Three identical uniform rods are each acted on by two or more forces, all perpendicular to the rods and all equal in magnitude. Which of the rods could be in static equilibrium if an additional force is applied at the center of mass of the rod?



(1) Only 1 (2) Only 2
(3) Only 3 (4) Only 1 and 2

19. Two spheres of masses m and M are situated in air and the gravitational force between them is F . The space around the masses is now filled with a liquid of specific gravity 3, the gravitational force will be now

(1) F (2) $\frac{F}{3}$ (3) $\frac{F}{9}$ (4) $3F$

20. The density of a newly discovered planet is twice that of earth. The acceleration due to gravity at the surface of the planet is equal to that at the surface of the earth. If the radius of the earth is R , the radius of the planet would be

(1) $4R$ (2) $\frac{1}{4}R$
(3) $\frac{1}{2}R$ (4) $2R$

21. If $\vec{r}$ is the force acting on a particle having position vector $\vec{r}$ and $\vec{\tau}$ be the torque of this force about the origin, then :

(1) $\vec{r} \cdot \vec{\tau} = 0$
(2) $\vec{r} \times \vec{\tau} = 0$
(3) $\vec{r} \cdot \vec{\tau} = r\tau$
(4) $\vec{r} \times \vec{\tau} = r\tau$

22. Let g be acceleration due to the gravity on the earth's surface. The gain in the potential energy of an object of mass m raised from surface of earth to a height equal to radius R of the earth is

(1) $\frac{1}{2}mgR$ (2) $2mgR$
(3) mgR (4) $\frac{1}{4}mgR$

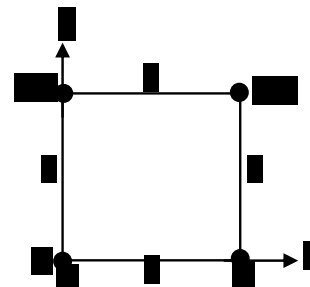
23. A fly wheel rotating a fixed axis has a kinetic energy of 360 J , when its angular speed is $30 \text{ radians per second}$. The moment of inertia of the flywheel about the axis of rotation is

(1) 0.15 kg m^2 (2) 0.75 kg m^2
(3) 0.60 kg m^2 (4) 0.80 kg m^2

24. Initial angular velocity of a circular disc of mass M is ω_1 . Then two small spheres of mass m are attached gently to diametrically opposite points on the edge of the disc. What is the final angular velocity of the disc?

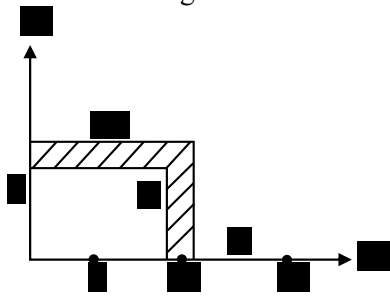
(1) $\left(\frac{M}{M+4m}\right)\omega_1$ (2) $\left(\frac{M+m}{m}\right)\omega_1$
(3) $\left(\frac{M+m}{M}\right)\omega_1$ (4) $\left(\frac{M}{M+2m}\right)\omega_1$

25. Four particles of masses $m, m, 2m$ and $2m$ are placed at the four corners of a square of side ' a ' as shown in figure. The (x, y) coordinates of the centre of mass are:

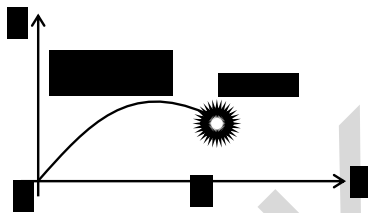


(1) $\left(\frac{a}{2}, 2a\right)$ (2) $\left(\frac{a}{2}, a\right)$
(3) $\left(\frac{a}{2}, \frac{2a}{3}\right)$ (4) $\left(a, \frac{a}{3}\right)$

26. The position vector of the centre of mass r_{cm} of an asymmetric uniform bar of negligible area of cross-section as shown in figure is



- (1) $\frac{5}{8}L\hat{y}$
 (2) $\frac{3}{8}L\hat{y}$
 (3) $\frac{11}{8}L\hat{y}$
 (4) $\frac{13}{8}L\hat{y}$
27. A projectile is fired at an angle and it was following a parabolic path. Suddenly, it explodes into fragments. Choose the correct statement regarding this situation.



- (1) Due to explosion CM shifts upwards.
 (2) Due to explosion CM shifts downwards.
 (3) Due to explosion CM traces its path back to origin.
 (4) CM continues to move along same parabolic path.

28. Match the Column I with Column II.

	Column I		Column II
(A)	For translational Equilibrium	(p)	Mk^2
(B)	For rotational Equilibrium	(q)	Angular acceleration
(C)	Moment of inertial of a body	(r)	$\sum \vec{r} \times \vec{v}$
(D)	Torque is required to produce	(s)	$\sum \vec{r}$

- (1) A – p, B – q, C – r, D – s
 (2) A – q, B – r, C – s, D – p
 (3) A – r, B – q, C – p, D – s
 (4) A – r, B – s, C – p, D – q

29. Two discs of moments of inertia I_1 and I_2 about their respective axes, rotating with angular frequencies ω_1 and ω_2 respectively, are brought into contact face to face with their axes of rotation coincident. The angular frequency of the composite disc will be

- (1) $\frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$ (2) $\frac{I_2\omega_1 + I_1\omega_2}{I_1 + I_2}$
 (3) $\frac{I_1\omega_1 - I_2\omega_2}{I_1 - I_2}$ (4) $\frac{I_2\omega_1 - I_1\omega_2}{I_1 - I_2}$

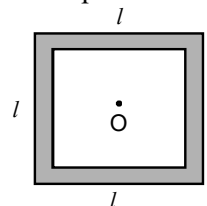
30. A child is standing with folded hands at the centre of a platform rotating about its central axis. The kinetic energy of the system is K . Now, the child stretches his arms so that moment of inertia of the system doubled. Now, the kinetic energy of the system is

- (1) $\frac{K}{4}$ (2) $\frac{K}{2}$
 (3) $2K$ (4) $4K$

31. A wheel of mass 5 kg and radius 0.40 m is rolling on a road without sliding with angular velocity 10 rad s^{-1} . The moment of inertia of the wheel about the axis of rotation is 0.65 kg m^2 . The percentage of kinetic energy of rotation to the total kinetic energy of the wheel is

- (1) 22.4% (2) 11.2%
 (3) 88.8 % (4) 44.8 %

32. Four thin rods of same mass M and same length l , form a square as shown in figure. Moment of inertia of this system about an axis through centre O and perpendicular to its plane is



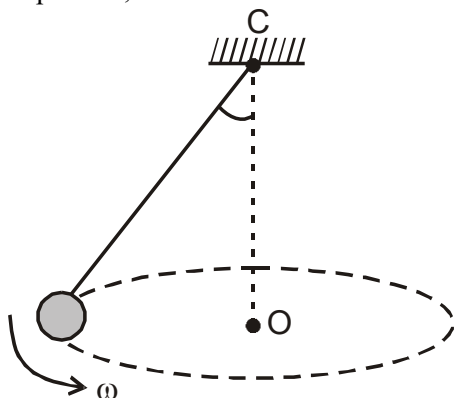
- (1) $\frac{4}{3}Ml^2$ (2) $\frac{Ml^2}{3}$
 (3) $\frac{Ml^2}{6}$ (4) $\frac{2}{3}Ml^2$

33. The ratio of the radii of gyration of a circular disc about a tangential axis in the plane of the disc and of a circular ring of the same radius about a tangential axis in the plane of the ring is

- (1) 2 : 3 (2) 2 : 1
 (3) $\sqrt{5} : \sqrt{6}$ (4) 1 : $\sqrt{2}$

34. A conical pendulum consists of a simple pendulum moving in a horizontal circle as shown. C is the

pivot, O the centre of the circle in which the pendulum bob moves and ω the constant angular velocity of the bob. If $\vec{L}$ is the angular momentum about point C, then



- (1) $\vec{L}$ is constant
- (2) only direction of $\vec{L}$ is constant
- (3) only magnitude of $\vec{L}$ is constant
- (4) none of the above.

35. If the equation for the displacement of a particle moving on a circular path is given by $(\theta) = 2t^3 + 0.5$, where θ is in radians and t in seconds, then the angular velocity of the particle after 2 sec from its start is

- (1) 8 rad/sec
- (2) 12 rad/sec
- (3) 24 rad/sec
- (4) 36 rad/sec

36. Assuming that the gravitational potential energy of an object at infinity is zero, the change in potential energy (final - initial) of an object of mass m , when taken to a height h from the surface of earth (of radius R) is given by,

- (1) $-\frac{GMm}{R+h}$
- (2) $\frac{GMmh}{R(R+h)}$
- (3) mgh
- (4) $\frac{GMm}{R+h}$

37. The time period of a geo-stationary satellite is 24 h, at a height $6R_E$ (R_E is the radius of earth) from surface of earth. The time period of another satellite whose height is $2.5R_E$ from surface of earth will be

- (1) $6\sqrt{2} h$
- (2) $12\sqrt{2} h$
- (3) $\frac{24}{25} h$
- (4) $\frac{12}{25} h$

38. The ratio of escape velocity at earth (v_e) to the escape velocity at a planet (v_p) whose radius and mean density are twice as that of earth is

- (1) $1:2\sqrt{2}$
- (2) $1:4$
- (3) $1:\sqrt{2}$
- (4) $1:2$

39. **Assertion (1):** If we move from poles to equator, the direction of acceleration due to gravity of earth always points towards the center of earth without any variation in its magnitude.

Reason (R): At equator, the direction of acceleration due to the gravity is towards the center of earth.

In the light of above statements, choose the correct answer from the options given below

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is NOT the correct explanation of A.
- (3) A is true but R is false
- (4) A is false but R is true

40. The orbital velocity of an artificial satellite in a circular orbit above the earth surface is v . For a satellite orbiting at an altitude half of the earth's radius, the orbital velocity is

- (1) $\frac{2v}{3}$
- (2) $\sqrt{\frac{2}{3}} v$
- (3) $\sqrt{\frac{3}{2}} v$
- (4) $\frac{3v}{2}$

41. Two bodies of mass m and $9m$ are placed at a distance R . The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (G = gravitational constant)

- (1) $-\frac{12Gm}{R}$
- (2) $-\frac{16Gm}{R}$
- (3) $-\frac{20Gm}{R}$
- (4) $-\frac{8Gm}{R}$

42. A satellite of the earth is revolving in a circular orbit with a uniform speed v . If the gravitational force suddenly disappears, the satellite will

- (1) Continue to move with velocity v along the original orbit
- (2) Move with a velocity v , tangentially to the original orbit
- (3) Fall down with increasing velocity
- (4) Ultimately come to rest somewhere on the original orbit

43. Match the column I with column II.

(where M_E is the mass of the earth, m is mass of the satellite and r is the radius of the orbit)

Column I	Column II
----------	-----------

(A)	Kinetic energy	(p)	$-\frac{GM_E m}{2r}$
(B)	Potential energy	(q)	$\sqrt{\frac{GM_E}{r}}$
(C)	Total energy	(r)	$-\frac{GM_E m}{r}$
(D)	Orbital velocity	(s)	$\frac{GM_E m}{2r}$

(1) A – r, B – s, C – q, D – P

(2) A – q, B – p, C – r, D – s

(3) A – p, B – q, C – s, D – r

(4) A – S, B – r, C – p, D – q

44. In case of an orbiting satellite, if the radius of orbit is decreased,

(1) its kinetic energy decreases

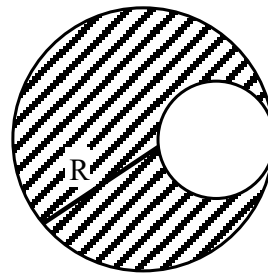
(2) its potential energy increases

(3) Both (1) and (2) are correct

(4) Both (1) and (2) are incorrect

45. A spherical hole is made in a solid sphere of radius R. The mass of the sphere before

hollowing was M. The gravitational field at the centre of the hole due to the remaining mass is – Fig.



- (1) Zero (2) $\frac{GM}{8R^2}$
 (3) $\frac{GM}{2R^2}$ (4) $\frac{GM}{R^2}$

CHEMISTRY

46. 2 moles of PCl_5 when heated in a closed vessels of 2 litre capacity at equilibrium, 40% of PCl_5 dissociated in PCl_3 and Cl_2 .

The value of the equilibrium constant is :

- (1) 2.67 (2) 5.3 (3) 0.533 (4) 0.267

47. The following reaction is favourable at –
 $\text{H}_2\text{O(s)} \rightleftharpoons \text{H}_2\text{O(l)} \quad \Delta H = +ve ; T = 0^\circ\text{C}$

- (1) Low pressure & low temperature
 (2) High pressure & high temperature
 (3) Low pressure & high temperature
 (4) None of these

48. Match the following :

Column-I (Assume only reactant were present initially)		Column- II	
(A)	For the equilibrium $\text{NH}_4\text{I(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{HI(g)}$, if pressure is increased at equilibrium	(p)	Forward shift
(B)	For the equilibrium $\text{H}_2\text{O(g)} + \text{CO(g)} \rightleftharpoons \text{H}_2\text{(g)} + \text{CO}_2\text{(g)}$, inert gas is added at constant pressure at equilibrium	(q)	No shift in equilibrium
(C)	For the equilibrium $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$, Cl_2 is removed at equilibrium	(r)	Backward shift

(1) (A - r) ; (B - q) ; (C - p)

(2) (A - q) ; (B - r) ; (C - p)

(3) (A - p) ; (B - q) ; (C - r)

(4) (A - p) ; (B - r) ; (C - q)

49. **Assertion :** A catalyst does not influences the values of equilibrium constant.

Reason : Catalysts influence the rate of both forward and backward reactions equally.

- (1) Both assertion and reason are correct, and the reason is the correct explanation for the assertion.
 (2) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion.
 (3) The assertion is incorrect, but the reason is correct.
 (4) Both are assertion and reason are incorrect.

50. The dissociation constant of two acids HA_1 and HA_2 are 3.14×10^{-4} and 1.96×10^{-5} respectively. The relative strength of the acids will be approximately :

- (1) 1 : 4 (2) 4 : 1
 (3) 1 : 16 (4) 16 : 1

51. When $\text{Cr(s)} + \text{OH}^-\text{(aq)} \rightarrow \text{Cr(OH)}_4^-\text{(aq)} + \text{H}_2\text{(g)}$ (basic solution) is balanced, the sum of the coefficients of all the reactants and products is:

- (1) 14 (2) 15 (3) 17 (4) 9

52. Which of the following statements is/are true if 1mol of H_3PO_x is completely neutralized by 40g of NaOH ?

- (1) $x = 2$ and acid is monobasic
 (2) $x = 3$ and acid is dibasic
 (3) $x = 3$ and acid is tribasic
 (4) $x = 4$ and given acid forms three series of salt.

53. Which of the following is/are redox reaction(s)?

- (1) $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 2\text{H}_2\text{O} + 3\text{S}$
 (2) $2\text{BaO} + \text{O}_2 \rightarrow 2\text{BaO}_2$
 (3) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
 (4) All are correct

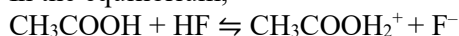
54. Which one is correctly matched?

Substance	O.N. of S
(1) H_2S	+2
(2) H_2SO_5	+4
(3) H_2SO_4	+4
(4) $\text{Na}_2\text{S}_4\text{O}_6$	+2.5

55. Solution of 0.01M MgCl_2 will form precipitate of Mg(OH)_2 at limiting pH of: Given K_{sp} of Mg(OH)_2 is 1×10^{-12}

- (1) 9 (2) 5 (3) 12 (4) 2

56. In the equilibrium,



- (1) F^- is the conjugate acid of CH_3COOH
 (2) F^- is the conjugate base of HF
 (3) CH_3COOH is the conjugate acid of $\text{CH}_3\text{COOH}_2^+$
 (4) $\text{CH}_3\text{COOH}_2^+$ is the conjugate base of CH_3COOH

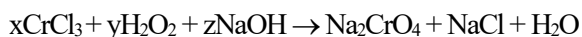
57. Which of the following reactions are disproportionation reaction ?

- (a) $2\text{Cu}^+ \longrightarrow \text{Cu}^{2+} + \text{Cu}^0$
 (b) $3\text{MnO}_4^{2-} + 4\text{H}^+ \longrightarrow 2\text{MnO}_4^- + \text{MnO}_2 + 2\text{H}_2\text{O}$
 (c) $2\text{KMnO}_4 \xrightarrow{\Delta} \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$
 (d) $2\text{MnO}_4^- + 3\text{Mn}^{2+} + 2\text{H}_2\text{O} \xrightarrow{\Delta} 5\text{MnO}_2 + 4\text{H}^+$

Select the correct option from the following :-

- (1) (a) and (b) only (2) (a), (b) and (c)
 (3) (a), (c) and (d) (4) (a) and (d) only

58. In the redox reaction



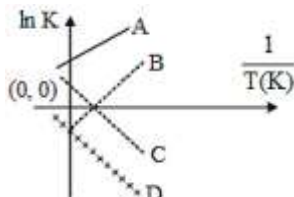
- (1) $x = 2, y = 4, z = 10$
 (2) $x = 2, y = 6, z = 5$
 (3) $x = 2, y = 3, z = 8$
 (4) $x = 2, y = 3, z = 10$

59. For the reaction,
 $I_2(g) \rightleftharpoons 2I(g)$; $K_C = 37.6 \times 10^{-6}$
 at 1000 K. If 1.0 mole of I is introduced into a 1.0 L flask at 1000 K, at equilibrium
 (1) $[I_2] > [I]$ (2) $[I_2] = [I]$
 (3) $[I_2] < [I]$ (4) Unpredictable
60. Consider the reaction where $k_p = 0.512$ at 500 K
 $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
 If the three gases are mixed in a rigid container so that the partial pressure of each gas is initially 2 atm, which is true?
 (1) More PCl_5 will be produced
 (2) More PCl_3 will be produced
 (3) Equilibrium will be established when 50% reaction is complete
 (4) None of the above
61. pH for the solution of salt undergoing anionic hydrolysis (say $CHCOONa$) is given by:
 (1) $pH = \frac{1}{2}[pK_w + pK_a + \log c]$
 (2) $pH = \frac{1}{2}[pK_w + pK_a - \log c]$
 (3) $pH = \frac{1}{2}[pK_w + pK_b - \log c]$
 (4) None of these
62. What weight of HNO_3 is needed to convert 5 g I_2 into HIO_3 ? (Assume HNO_3 converts into NO)
 (1) 4.13 g (2) 24.8 g
 (3) 6.2 g (4) 10.2 g
63. Which of the following is redox reaction?
 (1) $N_2O_5 + H_2O \rightarrow 2HNO_3$
 (2) $AgNO_3 + KI \rightarrow AgI + KNO_3$
 (3) $BaO + H_2SO_4 \rightarrow BaSO_4 + H_2O$
 (4) $SnCl_2 + HgCl_2 \rightarrow SnCl_4 + Hg$
64. In acidic medium and basic (strongly basic) medium, n-factor of $KMnO_4$ is x and y respectively. If n-factor of $K_2Cr_2O_7$ in acidic medium is z then the value of $x + y + z$ is
 (1) 14 (2) 12 (3) 10 (4) 9
65. The oxidation number of nickel in $K_4[Ni(CN)_4]$ is-
 (1) +1 (2) +2 (3) -1 (4) 0
66. The number of electrons lost or gained during the change $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$ is
 (1) 2 (2) 4 (3) 6 (4) 8
67. Which of the following combination on mixing in equal volumes will produce a buffer solution?
 A = 1 M NH_4Cl B = 1 M CH_3COONa
 C = 1 M NH_4OH D = 1 M CH_3COOH
 (1) C and D (2) A and B
 (3) A and C (4) B and C
68. Equal volumes of three acid solutions of pH 3, 4 and 5 are mixed in a vessel. What will be the H^+ ion concentration in the mixture?
 (1) $3.7 \times 10^{-3} M$ (2) $1.11 \times 10^{-3} M$
 (3) $1.11 \times 10^{-4} M$ (4) $3.7 \times 10^{-4} M$
69. To completely neutralize 0.135 g H_2AO_4 , 2.75×10^{-3} g equivalents of $NaOH$ is required. What will be the number of neutrons in A (Consider that number of neutrons and protons are equal in A)?
 (1) 16 (2) 12 (3) 9 (4) 32
70. The oxidation number of Hydrogen in KH , MgH_2 and NaH respectively would be :
 (1) +1, +1 and +1 (2) -1, -1 and -1
 (3) +2, +1 and -2 (4) -2, -3 and -1
71. Oxidation numbers of 'Fe' in Fe_3O_4 are
 (1) +2 and +3 (2) +1 and +2
 (3) +1 and +3 (4) +3 and +3
72. The equivalent weight of $KMnO_4$ in acidic and neutral medium are $\frac{M}{a}$ and $\frac{M}{b}$ respectively, where M is the molar mass of $KMnO_4$. Value of 'a' and 'b' are
 (1) 1, 5 (2) 5, 1 (3) 5, 3 (4) 3, 5
73. In the following reaction,
 $M^{x+} + 2MnO_4^- \rightarrow MO_3^- + Mn^{2+} + \frac{1}{2}O_2$
 If one mole of permanganate ion oxidizes 2.5 moles of M^{x+} , then the value of x is
 (1) 5 (2) 3 (3) 4 (4) 2
74. Calculate the ratio of $\frac{HCOONa}{HCOOH}$ in a buffer of pH 4.25. (K_a of $HCOOH = 1.8 \times 10^{-4}$)
 Given antilog (0.51) = 3.24
 (1) 5.00 (2) 4.75 (3) 3.24 (4) 5.69
75. The reaction, $3ClO^- (aq) \rightarrow ClO_3^- (aq) + 2Cl^- (aq)$ is an example of
 (1) oxidation
 (2) reduction
 (3) disproportionation
 (4) decomposition
76. The solubility product of CuS , Ag_2S and HgS are 10^{-31} , 10^{-44} and 10^{-54} respectively. The solubilities of these sulphides are in the increasing order
 (1) $CuS < Ag_2S < HgS$
 (2) $HgS < CuS < Ag_2S$
 (3) $Ag_2S < HgS < CuS$
 (4) $Ag_2S < CuS < HgS$

77. The ionization constant of ammonium hydroxide is 1.77×10^{-5} at 298 K. Hydrolysis constant of ammonium chloride is:

- (1) 6.5×10^{-12} (2) 5.65×10^{-13}
(3) 5.65×10^{-12} (4) 5.65×10^{-10}

78. Which of the following lines correctly show the temperature dependence of equilibrium constant K, for an exothermic reaction?



- (1) A and B (2) B and C
(3) C and D (4) A and D

79. The pH of 20 mL of 0.1 N HCl will not alter after the addition of

- (1) 20 mL of distilled water
(2) 1 mL of 0.1 N NaOH
(3) 1 mL of 1 N HCl
(4) 500 mL of 0.1N HCl

80. CH_3NH_2 (0.12 mole, $\text{pK}_b = 3.3$) is added to 0.08 moles of HCl and the solution is diluted to one litre, resulting pH of solution is.

- (1) 10.7 (2) 3.6
(3) 10.4 (4) 11.3

81. Which of the following cannot act both as Bronsted acid and as Bronsted base?

- (1) HSO_4^- (2) HCO_3^-
(3) NH_3 (4) HCl

82. Which of the following reactions is an example of a redox reaction?

- (1) $\text{XeF}_2 + \text{PF}_5 \rightarrow [\text{XeF}]^+ \text{PF}_6^-$
(2) $\text{XeF}_6 + \text{H}_2\text{O} \rightarrow \text{XeOF}_4 + 2\text{HF}$
(3) $\text{XeF}_6 + 2\text{H}_2\text{O} \rightarrow \text{XeO}_2\text{F}_4 + 4\text{HF}$
(4) $\text{XeF}_4 + \text{O}_2\text{F}_2 \rightarrow \text{XeF}_6 + \text{O}_2$

83. If 1 M KMnO_4 and 1 M $\text{K}_2\text{Cr}_2\text{O}_7$ solutions can oxidize Fe to Fe in acidic medium, then volume of oxidant required for 1 mol of Fe will be

- (1) $V_{\text{KMnO}_4} > V_{\text{K}_2\text{Cr}_2\text{O}_7}$
(2) $V_{\text{KMnO}_4} < V_{\text{K}_2\text{Cr}_2\text{O}_7}$
(3) $V_{\text{KMnO}_4} = V_{\text{K}_2\text{Cr}_2\text{O}_7}$
(4) cannot be predicted

84. Correct relation between dissociation constants of a dibasic acid is-

- (1) $k_{a_1} = k_{a_2}$ (2) $k_{a_1} > k_{a_2}$
(3) $k_{a_1} < k_{a_2}$ (4) $k_{a_1} = \frac{1}{k_{a_2}}$

85. Aqueous solution of SO_2 reacts with H_2S to precipitate sulphur, here SO_2 acts as

- (1) Catalyst
(2) Reducing agent
(3) Oxidizing agent
(4) Acid

86. The acidity and alkalinity of buffer solutions are constant because

- (1) These give unionized acid or base on reaction with added acid or alkali.
(2) Acids and alkalies in these solution are shielded from attack by other ions.
(3) They have excess of H^+ or OH^- ions.
(4) All of these

87. In hydrolysis of a salt of weak acid and strong base, $\text{A}^- + \text{H}_2\text{O} \rightleftharpoons \text{HA} + \text{OH}^-$, the hydrolysis constant (K_h) is equal to

- (1) $\frac{K_w}{K_a}$ (2) $\frac{K_w}{K_b}$
(3) $\sqrt{\frac{K_w}{C}}$ (4) $\frac{K_w}{K_a \times K_b}$

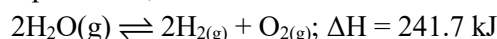
88. Among NH_3 , HNO_3 , NaN_3 and Mg_3N_2 , the number of molecules having nitrogen in negative oxidation state is –

- (1) 1 (2) 2 (3) 3 (4) 4

89. Which the pressure is applied over system ice $\rightleftharpoons$ water what will happen

- (1) More water will form
(2) More ice will form
(3) There will be no effect over equilibrium
(4) Water will decompose in H_2 and O_2

90. Consider the following reversible reaction at equilibrium,



Which one of the following changes in conditions will lead to maximum decomposition of $\text{H}_2\text{O}(\text{g})$?

- (1) Increasing both temperature and pressure.
(2) Increasing temperature and decreasing pressure.
(3) Decreasing both temperature and pressure.
(4) Keeping temperature and pressure constant.

BIOLOGY

91. **Assertion:** Arachidonic acid is an unsaturated fatty acid.

Reason: It contains carbon-oxygen double bond.

- (1) If both Assertion and Reason are true and the Reason is a correct explanation of Assertion.
- (2) If both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is true but the Reason is False.
- (4) If both Assertion and Reason are False.

92. Which of the following is an essential amino acid?

- (1) Phenylalanine (2) Alanine
- (3) Tyrosine (4) Glycine

93. The second meiotic division leads to

- (1) Separation of sex chromosomes
- (2) New DNA synthesis
- (3) Separation of chromatids and centromere
- (4) Separation of homologous chromosomes

94. Select the correct match.

- I. S-phase - DNA replication
 - II. Zygotene - Synapsis
 - III. Diplotene - Crossing over
 - IV. Meiosis - Both haploid and diploid cells
 - V. G_2 -phase - Quiescent stage
- (1) I and II only (2) III and IV only
 - (3) III and V only (4) I, III and V only

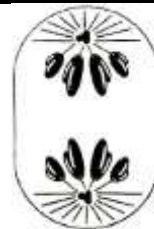
95. After performing the chemical analysis of organic compounds found in living organisms, two fractions were observed namely

- (1) Acid soluble pool and acid insoluble pool
- (2) Carbon pool and hydrogen pool
- (3) Inorganic pool and organic pool
- (4) Aqueous pool and non-aqueous pool

96. Based on the nature of R group there are many amino acids. If the R group in a proteinaceous amino acid is a methyl group, then the amino acid is:

- (1) Glycine (2) Alanine
- (3) Serine (4) Phenylalanine

97. Given diagram indicates which of the following phase of mitosis?



- (1) Interphase (2) Prophase
- (3) Metaphase (4) Anaphase

98. Which is correct w.r.t cilia?

- (1) Cilia are short hair-like outgrowths of the cell membrane.
- (2) Cilia cause the movement of either the cell or the surrounding fluid.
- (3) Cilia emerge from centriole like structure called the basal bodies.
- (4) All of the above

99. Which of the following statements is incorrect?

- (1) Lipids are strictly macromolecules
- (2) Palmitic acid has 16 carbons including carboxyl carbon
- (3) Oils have low melting point and hence remain as liquid.
- (4) Arachidonic acid is an unsaturated fatty acid

100. The differences in chemical and physical properties of amino acids are essentially of the :-

- (1) Amino group (2) Carboxyl group
- (3) R group (4) All of the above

101. How the presence of an enzyme affects the activation energy?

- (1) It is increased.
- (2) It is decreased.
- (3) It is first increased and then decreased.
- (4) Activation energy is not affected at all.

102. Which of the following amino acids has an amino group in its side chain?

- (1) Glycine (2) Alanine
- (3) Serine (4) Lysine

103. Upon complete hydrolysis, one molecule of triglyceride will yield how many molecules of fatty acids?

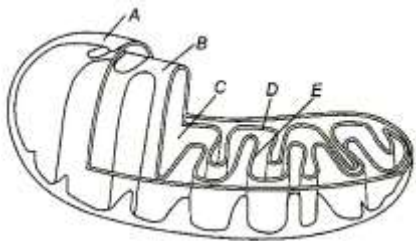
- (1) 1
- (2) 2
- (3) 3
- (4) Any of these is possible

- 104.** Which of the following statement is false regarding proteins?
- (1) A protein is heteropolymer and not a homopolymer
 - (2) Collagen is the most abundant protein in the animal world
 - (3) RUBISCO is the most abundant protein in the whole biosphere
 - (4) The first amino acid in the polypeptide chain is called as C-terminal amino acid and the last amino acid is called as N-terminal amino acid
- 105.** I. Chromosomes cluster at opposite poles loss their identity as discrete elements.
II. Nuclear envelope assembles around the chromosome clusters.
III. Nucleolus, Golgi complex and ER reappear
Above features indicate which phase of mitosis?
- (1) Anaphase
 - (2) Telophase
 - (3) Cytokinesis
 - (4) S-phase
- 106.** Read the following statements and identify the option given.
- I. In prokaryotic cell, the nuclear membrane, chloroplast, mitochondria, microtubules and different kinds of pili are absent.
 - II. In eukaryotic cell, the nuclear membrane, chloroplast, mitochondria and pili are present.
 - III. In prokaryotic cell, the ribosome is of 70s type and in mitochondria of eukaryotic animal cell, the ribosome is 80s type
- (1) I and II are wrong; III is correct
 - (2) I is correct; II and III are wrong
 - (3) I and III are correct; III is wrong
 - (4) I, II and III are wrong
- 107.** An organism exclusively with 70s type of ribosomes contains which of the following
- (1) DNA enclosed within the nuclear membrane
 - (2) Circular naked DNA
 - (3) Double-stranded DNA with protein coat
 - (4) Single-stranded DNA with protein coat
- 108.** Small disc-shaped structures at the surface of the centromeres that appear during metaphase are
- (1) Kinetochore
 - (2) Metaphase plate
 - (3) Spindle fibres
 - (4) Chromatid
- 109.** which of the following r-RNAs acts as structural RNA as well as ribozyme in bacteria?
- (1) 23S rRNA
 - (2) 16S rRNA
 - (3) 28S rRNA
 - (4) 18S rRNA
- 110.** During which phase of mitotic cell cycle, chromosomal duplication takes place?
- (1) G1 phase
 - (2) G2 phase
 - (3) S phase
 - (4) Anaphase
- 111. Assertion :** Entire content of nucleolus is continuous with rest of the nucleoplasm.
Reason : Nucleolus is not covered by any type of membrane.
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 - (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 - (3) Assertion is True but the Reason is False.
 - (4) Both Assertion & Reason are False.
- 112.** Match the following columns
- | Column-I | | Column- II | |
|----------|-----------|------------|---------------------|
| (A) | Inulin | (i) | Hormone |
| (B) | Galactose | (ii) | Alkaloids |
| (C) | Morphine | (iii) | Sugar |
| (D) | Trypsin | (iv) | Enzyme |
| (E) | Insulin | (v) | Polymer of fructose |
- (1) A-(ii), B-(i), C-(iv), D-(iii), E-(v)
 - (2) A-(iv), B-(ii), C-(i), D-(iii), E-(v)
 - (3) A-(v), B-(iii), C-(ii), D-(iv), E-(i)
 - (4) A-(iv), B-(iii), C-(ii), D-(v), E-(i)
- 113.** Give the name of the phases of meiosis, in which
- I. The chromosome number is reduced to haploid state
 - II. The amount of DNA is reduced to half
- The correct option is
- (1) Anaphase-II; anaphase-I
 - (2) Anaphase-I, metaphase-II
 - (3) Anaphase-I, anaphase-II
 - (4) Anaphase-II, metaphase-I
- 114.** Read the following statements and choose the correct option.
- (1) The cisternae of Golgi are concentrically arranged near the nucleus with distinct concave cis or the forming face and convex trans or the maturing face.
 - (2) The Golgi cisternae are concentrically arranged near the nucleus with distinct convex cis or the maturing face and concave trans or the forming face.
 - (3) The Golgi apparatus principally performs the function of packaging materials, to be delivered either to the intracellular targets or secreted outside the cell.
 - (4) A number of proteins synthesised by ribosomes on the endoplasmic reticulum are modified in the cisternae of the Golgi apparatus before they are released from its cis face.

115. Assertion: pollen mother cells undergoes meiosis.
Reason: Each pollen mother cell produces four haploid pollen grains.

- (1) Assertion is true, Reason is true Reason is correct explanation for Assertion
- (2) Assertion is true, Reason is true Reason is not correct explanation for Assertion
- (3) Assertion is true, Reason is false
- (4) Assertion is false, Reason is true

116. Go through the sectional view of a mitochondrion showing the different parts and identify the structures A to E



- (1) A-Outer membrane, B-Inner membrane, C-Matrix, D-Inter membrane space, E-Crista
- (2) A-Outer membrane, B-Inner membrane, C-Inter-membrane space, D-Matrix, E-Crista
- (3) A-Outer membrane, B-Inner membrane, C-Matrix, D-Crista, E-Inter membrane space
- (4) A-Outer membrane, B-Inner membrane, C-Crista, D-Matrix, E-Inter-membrane space

117. I. It is the extension of plasma membrane into the cytoplasm

II. It helps in cell wall formation, DNA replication, respiration, secretion processes, increases the surface area of plasma membrane and enzymatic contents.

It also helps in cytokinesis

III. It is the characteristic of bacterial cells.

The above features are attributed to bacteria

- (1) Plasmid
- (2) Nucleoid
- (3) Mesosome
- (4) Pilus

118. Microtubular arrangement in axoneme of cilia is:-

- (1) 9 (Peripheral doublets) + 0
- (2) 9 (Peripheral doublets) + 2
- (3) 9 (Peripheral triplets) + 0
- (4) 9 (Peripheral triplets) + 2

119. Assertion: In animal cells, the cytokinesis is achieved by the appearance of a furrow in plasma membrane

Reason: In plant cells, the formation of the new cell wall begins with the formation of simple precursor called cell plate.

- (1) Assertion is true, Reason is true; Reason is correct explanation for Assertion
- (2) Assertion is true, Reason is true; Reason is not correct explanation for Assertion
- (3) Assertion is true, Reason is false
- (4) Assertion is false, Reason is true

120. The part of mitotic cycle occurring between two successive division phases is called as :-

- (1) Gap phase
- (2) Metaphase
- (3) Interphase
- (4) S-phase

121. Nuclear envelope is derived from :-

- (1) Membrane of Golgi complex
- (2) Microtubules
- (3) RER
- (4) SER

122. Assertion: Meiosis is known as reductional division

Reason: During meiosis, the chromosome number get reduced by half of its total number

- (1) Assertion is True, Reason is True Reason is correct explanation for Assertion
- (2) Assertion is True, Reason is True Reason is not correct explanation for Assertion
- (3) Assertion is True, Reason is False
- (4) Assertion is False, Reason is True

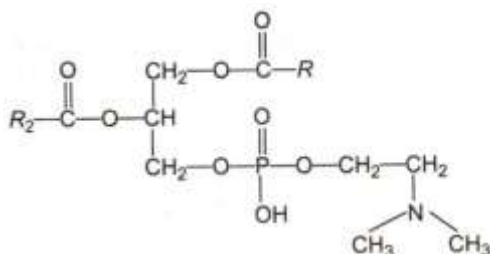
123. Which of the following statement(s) is/are not correct about meiosis?

- I. Meiosis involves pairing of homologous chromosomes and recombination between them
 - II. Two diploid cells are formed at the end of meiosis-II
 - III. Meiosis involves two sequential cycles of nuclear and cell division called meiosis-I and meiosis-II, but only a single cycle of DNA replication
 - IV. Meiosis-I is initiated after the parental chromosome duplication which produce identical sister chromatids at the S-phase
- (1) I and III
 - (2) II only
 - (3) II and III
 - (4) I, II, III and IV

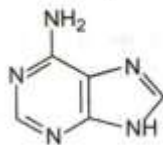
124. Lysosomal enzymes are optimally active at :-

- (1) Neutral pH (2) Basic pH
(3) Acidic pH (4) Any of these

125. Which one of the following structural formulae of two organic compounds is correctly identified along with its related function?



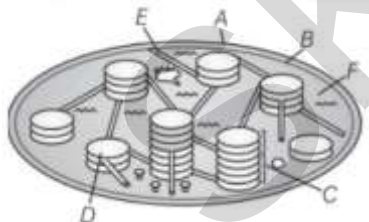
(A)



(B)

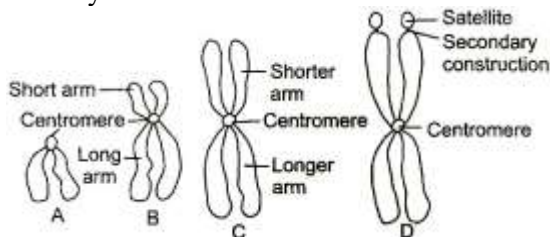
- (1) A-Triglyceride → major-Source of energy
(2) B- Uracil → A component of DNA
(3) A-Lecithin → A component of cell membrane
(4) B-Thymine → A nucleotide that makes up nucleic acids

126. Identify A to F in the sectional view of a chloroplast showing the different parts



- (1) A-Inner membrane, B-Granum, C-Outer membrane, D-Stroma lamella, E-Stroma, F-Thylakoid
(2) A-Outer membrane, B-Inner membrane, C-Granum, D-Thylakoid, E-Stroma lamella, F-Stroma
(3) A-Thylakoid, B-Outer membrane, C-Stroma, D-Stroma lamella, E-Granum, F-Inner membrane
(4) A-Outer membrane, B-Stroma, C-Inner membrane, D-Granum, E-Thylakoid, F-Stroma lamella

127. In the given diagram of types of chromosomes, identify A-D



- (1) A-Telocentric chromosome, B-Acrocentric chromosome, C-Submetacentric chromosome, D-Metacentric chromosome
(2) A-Acrocentric chromosome, B-Telocentric chromosome, C-Metacentric chromosome, D-Submetacentric chromosome
(3) A-Submetacentric chromosome, B-Metacentric chromosome, C-Telocentric chromosome, D-Acrocentric chromosome
(4) A-Metacentric chromosome, B-Submetacentric chromosome, C-Acrocentric chromosome, D-Telocentric chromosome

128. Which of the following features are correct regarding ribosomes?

- I. Non-membrane bound
II. Absent in plastids and mitochondria
III. Present in the cytoplasm and on surface of RER
IV. Take part in protein synthesis

The correct option is

- (1) Only II (2) I and II
(3) I, II, III and IV (4) I, III and IV

129. Which of the following feature is present in mitochondria?

- (1) Linear DNA, 70S ribosome
(2) Circular DNA, ds DNA, 70S ribosome
(3) Circular DNA, glycogen synthesis
(4) Circular DNA, single stranded DNA, 70S ribosome

130. Match the following column

Column-I		Column- II	
(A)	Oxidoreductase	(i)	Interconversion of optical, geometrical positional isomers
(B)	Ligases	(ii)	Group transfer
(C)	Isomerases	(iii)	Oxidoreduction between two substrates
(D)	Hydrolases	(iv)	Linking together of two compounds
(E)	Transferases	(v)	Hydrolysis

- (1) A-(v), B-(iv), C-(i), D-(ii), E-(iii)
(2) A-(iv), B-(i), C-(v), D-(ii), E-(iii)
(3) A-(v), B-(iv), C-(ii), D-(i), E-(iii)
(4) A-(iii), B-(iv), C-(i), D-(v), E-(ii)

131. Which of the following pairs is mismatched?
- Pili – involved in locomotion
 - Cell wall – protective, determines shape, prevents from bursting
 - Glycocalyx – may be capsule or slime layer
 - Flagella, pili and fimbriae – surface structures of bacterial cell

132. **Assertion:** Ribosomes are considered as the most primitive cell organelles.

Reason: Ribosomes occur only in cytoplasm of all eukaryotic organisms.

- Assertion is True, Reason is True; Reason is correct explanation for Assertion
- Assertion is True, Reason is True; Reason is not correct explanation for Assertion
- Assertion is True, Reason is False
- Assertion is False, Reason is True

133. Which of the following properties of starch makes it useful as storage material?

- Easily translocated
- Chemically non reactive
- Easily digested by animals
- Osmotically inactive
- Synthesized during photosynthesis

- i, iii, and v
- i and v
- ii and iii
- ii and iv

134. **Statement I:** Microbodies are membrane bound vesicles containing various enzymes that are present exclusively only in plant cells.

Statement II: Sometimes a few chromosomes have non-staining secondary constrictions at a variable location

- Only statement I is correct
- Only statement II is correct
- Both statements I and II are correct
- Both statements I and II are incorrect

135. Match the following and choose the correct combination from the option given below.

Column-I		Column- II	
(A)	Endoplasmic reticulum	(i)	Take part in cellular respiration
(B)	Free ribosome	(ii)	Take part in osmoregulation and excretion
(C)	Mitochondrion	(iii)	Synthesis of lipids
(D)	Contractile vacuole	(iv)	Synthesise non-secretory proteins

- A-(iii), B-(iv), C-(i), D-(ii)
- A-(i), B-(ii), C-(iv), D-(iii)
- A-(iii), B-(ii), C-(i), D-(iv)
- A-(iii), B-(ii), C-(iv), D-(i)

136. I. One end of a polysaccharide is called non-reducing end while other end is called reducing end
- II. Starch (amylopectin) and glycogen are branched molecules
- III. Starch and glycogen are the reserve food materials of plants and animals, respectively.
- IV. Starch can hold iodine molecules in its helical secondary structure but cellulose being non-helical, cannot hold iodine
- Statements I and II are incorrect
 - All statement are incorrect
 - Only statement IV is incorrect
 - All statements are correct

137. Read the following statements :-

- Presence of DNA
- Presence of cristae
- Presence of ribosomes
- Enzymes for carbohydrate synthesis
- Site for oxidative phosphorylation

How many of the above statements are common for both mitochondria and chloroplast?

- Five
- Four
- Two
- Three

138. The M-phase starts with the ...A..., corresponding to the separation of daughter chromosomes, known as ...B... and usually ends with division of cytoplasm which is known as ...C... Identify A-C

- A-cell division; B-cytokinesis; C-karyokinesis
- A-nuclear division; B-karyokinesis; C-cytokinesis
- A-cell division; B-karyokinesis; C-cytokinesis
- A-nuclear division; B-cytokinesis; C-karyokinesis

139. Arrange the steps of catalytic cycle of an enzyme in order and choose the correct option.

- The enzyme releases the products of the reaction and the enzyme is free for another substrate.
- The active site of enzyme is in close proximity of the substrate and breaks of chemical bonds of the substrate.
- The binding of substrate induces the enzyme to alter its shape fitting more tightly around the substrate.
- The substrate binds to the active site of the enzyme fitting into the active site.

- IV, III, II, I
- III, II, I, IV
- IV, II, I, III
- II, I, IV, III

140. Assertion: Mitosis is also called as equational division.

Reason: Progeny cells formed as the result of mitosis have equal number of chromosomes among themselves.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

141. Consider the following statements about plant cytokinesis

- I. It usually occurs by cell plate method
 - II. The spindle usually persists during cytokinesis
 - III. Cell wall formation starts in the centre of the cell.
- Which of the statements given above are correct?

- (1) I and II
- (2) I and III
- (3) II and III
- (4) I, II and III

142. Assertion: Mitochondria are known as power houses of the eukaryotic cells.

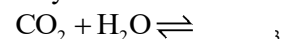
Reason: Mitochondria create energy.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

143. With reference to enzymes, which one of following statements is true?

- (1) Apoenzyme=Holoenzyme +Coenzyme
- (2) Holoenzyme=Apoenzyme +Coenzyme
- (3) Coenzyme=Apoenzyme +Holoenzyme
- (4) Holoenzyme=Coenzyme+ Co-factor

144. The following reaction is catalyzed by which of the enzyme?



- (1) Cytochrome oxidase
- (2) Carbonic anhydrase
- (3) Carbonic transferases
- (4) Isomerases

145. Which of the following is not an example of terminally differentiated cells?

- (1) Mature nerve cells
- (2) Mature muscle cells
- (3) Mature human RBCs
- (4) Mature hepatocytes

146. Which is correct for meiotic metaphase-I?

- (1) Bivalents are arranged at equator
- (2) Univalents are arranged at equator
- (3) Non-homologous chromosomes forms pair

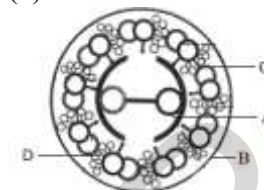
(4) Spindle fibres are attached at chromomere

147. Assertion: In the solutions of different pH, structure of amino acids changes

Reason: It is because of the ionizable nature of $-\text{NH}_2$ and $-\text{COOH}$ groups

- (1) Assertion is True, Reason is True; Reason is correct explanation for Assertion
- (2) Assertion is True, Reason is True; Reason is not correct explanation for Assertion
- (3) Assertion is True, Reason is False
- (4) Assertion is False, Reason is True

148.



Identify A, B, C and D :-

	(A)	(B)	(C)	(D)
(1)	Central sheath	Plasma membrane	Interdoublet bridge	Radial spoke
(2)	Central microtubule	Plasma membrane	Peripheral microtubule	Interdoublet bridge
(3)	Central sheath	Plasma membrane	Peripheral microtubule	Radial spoke
(4)	Central microtubule	Central sheath	Interdoublet bridge	Linker

149. Statement-I: All living organisms are composed of cells and products of cells.

Statement-II: All cells arise from pre- existing cells.

- (1) Statement-I and Statement-II both are correct
- (2) Statement-I is incorrect but Statement- II is correct
- (3) Statement-I and Statement-II both are incorrect
- (4) Statement-I is correct but Statement-II is incorrect

150. Select the secondary metabolites from the list given below

- | | |
|--------------|-----------------------|
| I. Alkaloids | II. Flavonoids |
| III. Rubber | IV. Essential oils |
| V. Morphine | VI. Coloured pigments |
| VII. Scents | VIII. Gums |
| IX. Spices | |

Choose the correct option

- (1) All I to IX
- (2) All except II and IX
- (3) I, III, IV and VI
- (4) All except I and VII

151. Which molecules present in the cell membrane limit its permeability for polar molecules ?

- (1) Carbohydrates
- (2) Phospholipids
- (3) Proteins
- (4) Water

- 152.** A cell organelle containing ribonucleoprotein
 (1) Lysosome (2) Microsome
 (3) Ribosome (4) Mesosome.
- 153.** Which of the following statements about the daughter cells formed as the result of mitotic cell division is correct?
 (1) They are genetically different from each other and from the parent cell
 (2) They are genetically identical with each other and with the parent cell
 (3) They are genetically identical with each other but different from the parent cell
 (4) Only one of the two daughter cells is genetically identical with the parent cell
- 154.** Read the following statements and choose the incorrect statement.
 (1) Once the chromatids reach the two poles, the chromosomal condensation starts, nucleolus and the nuclear membrane reappears.
 (2) In the equational division, the chromosome number of the parent is conserved in the daughter cell.
 (3) Meiosis occurs in the diploid cells to form gametes.
 (4) In sexual reproduction when the two gametes fuse the chromosome number is restored to the value in the parent.
- 155.** After the s-phase, what comprises a single chromosome?
 (1) Two daughter chromosomes
 (2) A double-stranded DNA molecule
 (3) Two single-stranded molecules of DNA
 (4) Two sister chromatids
- 156.** In animals, mitosis is not involved in :-
 (1) Wound healing
 (2) Replacement of lost blood cells
 (3) Formation of brain
 (4) Formation of gametes
- 157.** The anaphase II of meiosis II begins with the simultaneous splitting of the centromere of each chromosome allowing them to move towards opposite poles of the cell by
 (1) Shortening of microtubules attached to kinetochores
 (2) Shortening of microtubules attached to centromere
 (3) Cytokinesis followed by karyokinesis
 (4) Karyokinesis followed by cytokinesis
- 158.** Which cell organelle is concerned with synthesis of steroidal hormones?
 (1) RER (2) SER
 (3) Golgi complex (4) Lysosomes
- 159.** Analyse the following pairs and identify the correct option given.
 I. Chromoplasts – Contain pigments other than chlorophyll
 II. Leucoplasts – Devoid of any pigments
 III. Amyloplasts – Store proteins
 IV. Aleuroplasts – Store oils and fats
 V. Elaioplasts – Store carbohydrates
 (1) II and III are correct
 (2) III and IV are correct
 (3) IV and V are correct
 (4) I and II are correct
- 160.** Plasmodesmata are
 (1) Lignified, cemented between cells
 (2) Locomotory structures
 (3) Membranes connecting the nucleus with plasmalemma
 (4) Connections between adjacent cells
- 161.** Which of the following statements are correct?
 I. Human RBC is about 7.0 μm in diameter
 II. Cytoplasm is the main arena of cellular activities
 III. The shape of the cells may vary with the function they perform
 IV. Various chemical reactions occur in cytoplasm to keep the cell in the living state
 Choose the correct option
 (1) I, III and IV (2) I, IV and II
 (3) I, II, III and IV (4) II, III and IV
- 162.** Select the right option which relates to Schwann regarding the following statement
 I. He reported that cells had a thin outer layer which is today known as plasma membrane
 II. Cell wall is a unique character of the plant cells.
 III. Body of plants and animals are composed of cells and product of cells
 Choose the correct option given below
 (1) All are incorrect
 (2) Only III is correct
 (3) All are correct
 (4) Only II and III are correct
- 163.** Which of the following is the most abundant element in living cells ?
 (1) Oxygen (2) Carbon
 (3) Hydrogen (4) Nitrogen

164 Assertion: Mitochondria and chloroplast are semi-autonomous organelles.

Reason: They are formed by division of pre-existing organelles as well as contain DNA but lack protein synthesizing machinery.

- (1) Assertion is True, Reason is True Reason is correct explanation for Assertion
- (2) Assertion is True, Reason is True Reason is **not** correct explanation for Assertion
- (3) Assertion is True, Reason is False
- (4) Assertion is False, Reason is True

165. Read the following statements :-

- A. The ribosomes of the chloroplasts are smaller than the cytoplasmic ribosomes.
- B. Centriole are made of nine unevenly spaced peripheral fibrils of tubulin protein.
- C. Nuclear pores act as passage for proteins and RNA molecules in both directions between nucleus and cytoplasm.
- D. Centrosome and centromere form the basal body of cilia and flagella that facilitate locomotion.
- E. Lysosomes are non-membrane structures containing enzymes for digestion of all types of macromolecules.

How many statements are correct?

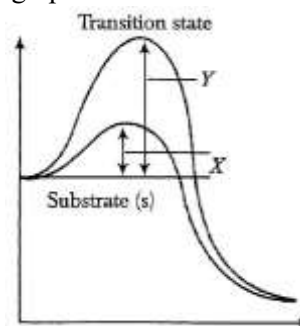
- (1) One
- (2) Three
- (3) Two
- (4) Five

166. Match the items in column I with column II and choose the correct option.

Column-I		Column- II	
(A)	Sap vacuole	(i)	Contain digestive enzyme
(B)	Contractile vacuule	(ii)	Store metabolic gases
(C)	Food vacuole	(iii)	Excretion
(D)	Gas vacuole	(iv)	Store and concentrate mineral salts and nutrients

- (1) A-(iv), B-(iii), C-(i), D-(ii)
- (2) A-(ii), B-(iii), C-(iv), D-(i)
- (3) A-(iv), B-(ii), C-(iii), D-(i)
- (4) A-(iv), B-(iii), C-(ii), D-(i)

167. Choose the correct option representing X and Y in the given graph



- (1) X-Activation energy without enzymes, Y-Activation energy with enzyme
- (2) X- Activation energy with enzyme, Y-Activation energy without enzyme
- (3) X-Substrate concentration with enzyme, Y-Substrate concentration without enzyme
- (4) X-Substrate concentration without enzyme, Y-Substrate concentration with enzyme

168. I. ...A... phase corresponds to the interval between mitosis and initiation of DNA replication

II. In animal cells, during the ...B... phase, DNA replication begins in the nucleus and the centriole duplicates in the cytoplasm

III. During the ...C... phase, proteins are synthesized for the preparation of mitosis, while cell growth continues

Identify the blanks (A-C) to complete the given statements (I-III).

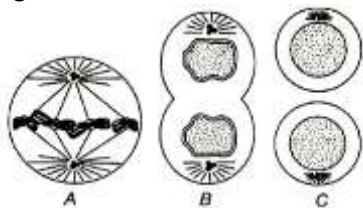
- (1) A-G₂, B-S, C-G₁
- (2) A-S, B-G₂, C-G₁
- (3) A-S, B-G₁, C-G₂
- (4) A-G₁, B-S, C-G₂

169. Choose the correct statements regarding cell cycle

- I. Interphase is called the resting phase
- II. Interphase is the time during which the cell is preparing for division
- III. The interphase is divided into phases, *i. e.*, G₁, S and G₂-phase
- IV. Interphase represents the phase between the two successive M-phases

- (1) I and IV Only
- (2) II and III Only
- (3) I and III Only
- (4) I, II, III and IV Only

170. See the diagrams carefully and identify the different stages of mitosis (A – C) by choosing appropriate options given below



- (1) A-Metaphase; B-Telophase; C-Interphase
 (2) A-Telophase; B-Metaphase; C-Prophase
 (3) A-Anaphase; B-Telophase; C-Interphase
 (4) A-Telophase; B-Anaphase; C-Prophase
171. Select the correct statements regarding S-phase of interphase
- Occurs between G_1 and G_2
 - DNA replication begins in the nucleus
 - Centrioles duplicate in the cytoplasm
 - As DNA is doubled, number of chromosomes also doubles
- The option with correct statements is
- (1) IV and III (2) I, II, III and IV
 (3) II, III and IV (4) I, II and III
172. Choose the **incorrect** statement with respect to cell membrane :-
- Cell membrane is an asymmetric structure
 - Cholesterol controls fluidity of animal cell membrane
 - Phospholipid molecules frequently exhibit flip-flop movement
 - Integral proteins can not be easily detached from cell membrane

Ans. (3)

173. Match the columns and find out the correct combination:

Column-I		Column-II	
A	DNA duplication	i	Anaphase
B	Double the number of chromosomes	ii	Cytokinesis
C	Double the number of cells	iii	S-phase
D	Synthesis of some special protein.	iv	G_2 -phase

- (1) A-(iii), B-(i), C-(ii), D-(iv)
 (2) A-(i), B-(iii), C-(iv), D-(ii)
 (3) A-(iv), B-(ii), C-(i), D-(iii)
 (4) A-(ii), B-(iv), C-(iii), D-(i)

174. Find the correctly matched pairs and choose the correct option

- Leptotene – The chromosomes become invisible
 - Zygotene – Pairing of homologous chromosomes
 - Pachytene – Dissolution of the synaptonemal complex takes place
 - Diplotene – Bivalent chromosomes appear as tetrads
 - Diakinesis – Terminalisation of chiasmata takes place
- (1) I and II (2) II and IV
 (3) II and V (4) II and III

175. Match the following column

Column-I		Column- II	
(A)	Prosthetic group	(i)	Coordination bond with apoenzyme
(B)	Coenzyme	(ii)	Tightly bound to apoenzyme
(C)	Metal ions	(iii)	Transient binding with apoenzyme

- (1) A-(ii), B-(iii), C-(i)
 (2) A-(iii), B-(ii), C-(i)
 (3) A-(ii), B-(i), C-(iii)
 (4) A-(i), B-(ii), C-(iii)

176. Choose the correct combination of options to select the correct statement for prophase

- Chromosomal material condenses to form compact chromosomes
 - The assembly of mitotic spindle is initiated by the microtubules
 - Cells do not show organelles when viewed under the microscope in prophase
 - The nucleolus or nucleoli disappear completely
- (1) I only (2) II and III only
 (3) I and II only (4) All of these

177. Three of the following statements regarding cell organelles are correct, while one is wrong. Which one is wrong?

- Lysosomes are double membraned vesicles budded off from Golgi apparatus and contain digestive enzymes
- Endoplasmic reticulum consists of a network of membranous tubule and helps in transport, synthesis and secretion
- Leucoplasts are bound by two membranes, lack pigment but contain their own DNA and protein synthesizing machinery
- Mitochondria is double membrane bound structure that divide by fission.

178. Assertion: Meiotic division occurs in reproductive cells

Reason: Synapsis occurs during zygotene of meiosis I.

- (1) Assertion is True, Reason is True Reason **is** correct explanation for Assertion
- (2) Assertion is True, Reason is True; Reason **is not** correct explanation for Assertion
- (3) Assertion is True, Reason is False
- (4) Assertion is False, Reason is True

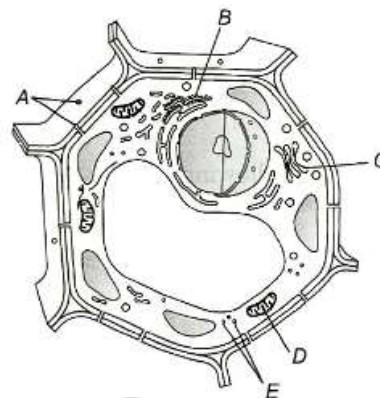
179. Which of the following statements are correct?

- I. Occurrence of different types of tissues, organs and organ system results in division of labour
- II. A new cell always develops by the division of a pre-existing cells
- III. Some cells are totipotent
- IV. Cell is the smallest unit, capable of independent existence and performing the essential functions of life

The correct option is

- (1) I, II and III only
- (2) I, III and IV only
- (3) II, III and IV only
- (4) I, II, III and IV

180. The following diagram shows some of the missing structures in a plant cell (A-E). Identify the structures



- (1) A-Plasmodesmata, B-Rough endoplasmic reticulum, C-Golgi apparatus, D-Mitochondrion, E-Ribosomes
- (2) A-Desmosome, B-Rough endoplasmic reticulum, C-Golgi apparatus, D-Mitochondrion, E-Ribosomes
- (3) A-Plasmodesmata, B-Smooth endoplasmic reticulum, C-Golgi apparatus, D-Mitochondrion, E-Ribosomes
- (4) A-Desmosome, B-Spherosome, C-Golgi apparatus, D-Mitochondrion, E-Ribosomes

Answer Key and Solution

ANSWER KEY & SOLUTION

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	3	1	4	4	1	1	1	4	1	4	1	4	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	1	3	1	3	3	1	4	1	3	1	4	4	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	1	3	3	3	2	1	1	4	2	2	2	4	4	3
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	2	1	1	2	2	1	4	4	1	2	1	4	1	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	1	1	4	2	4	4	3	4	1	2	1	3	2	3	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	4	1	4	3	4	4	1	2	3	1	1	3	1	2
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	2	1	3	1	1	2	4	4	1	3	2	4	3	4	2
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	4	2	1	1	3	1	3	3	3	1	1	3	2	2	3
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	3	1	2	3	3	2	1	4	2	4	1	3	4	4	1
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	4	3	2	1	2	4	3	2	2	4	1	1	2	1	1
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	2	3	2	1	4	4	1	2	4	4	3	3	1	3	3
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	1	2	4	4	1	4	3	1	3	1	4	1	2	4	1

PHYSICS

1. (1)

2. (1)

Sol. $T = 2\pi\sqrt{\left(\frac{\ell}{g}\right)}$. When g is doubled, for T to be constant, ℓ should be doubled.

3. (3)

Sol. The g decreases both we go below or above the surface of the earth.

4. (1)

Sol. $g_h = g\left[1 - \left(\frac{2h}{R}\right)\right] = g\left[1 - \left(\frac{d}{R}\right)\right] = g_d$
 $\Rightarrow d = 2h$

5. (4)

Sol. $g = \frac{GM}{R^2}$ (1)

$$g' = \frac{GM}{\left(\frac{90}{100}R\right)^2} = \frac{100}{81} \frac{GM}{R^2} \text{ (2)}$$

From equation (1) and (2)

$$g' = \frac{100}{81}g \Rightarrow \frac{g'}{g} = \frac{100}{81}$$

$$\Rightarrow \frac{g'}{g} = \frac{100}{81} \Rightarrow \frac{g'}{g} - 1 = \frac{100}{81} - 1$$

$$\frac{g' - g}{g} = \frac{19}{81}$$

$$\therefore \frac{\Delta g}{g} \times 100 = \frac{19}{81} \times 100 \approx$$

So increase is more than 19% of g .

6. (4)

Escape velocity is independent of angle of projection.

7. (4) 11 km/s

Sol. $= v_e = \sqrt{2gR} = \sqrt{2 \times 10 \times 6400 \times 10^3} \approx$ s

8. (1)

Sol. The velocity of the mass while reaching the surface of both the planets will be same.

$$\text{i.e., } \sqrt{2g'h'} = \sqrt{2gh}$$

$$\sqrt{2 \times g \times h'} = \sqrt{2 \times 9g \times 2}$$

$$\Rightarrow 2h' = 36 \Rightarrow h' = 18 \text{ m.}$$

9. (1)

Sol. Inside hollow sphere potential is same as that of surface.

$$\text{So, } V_A = V_B = V_O$$

10. (4)

Sol. Applying the theorem of perpendicular axis
 $I_d = I_c/2 = 200g \text{ cm}^2/2 = 100g \text{ cm}^2$

11. (1)

Sol. $\tau = dL/dt = (4A_0 - A_0)/4 = 3A_0/4.$

12. (4)

Sol. $V_{\text{escape}} = \sqrt{2gR}$

$$\frac{v_1}{v_2} = \sqrt{\frac{g_1 R_1}{g_2 R_2}} = \sqrt{pq}$$

13. (1)

Sol. kinetic energy $= \frac{1}{2} I\omega^2$

and for ring $I = mr^2$

$$\text{Hence KE} = \frac{1}{2} mr^2 \omega^2$$

14. (4)

Sol. COM lies closer to heavier mass
 t = thickness of plates

$$\text{Mass of circular plate } m_1 = \rho\pi\left(\frac{d}{2}\right)^2 t$$

$$\Rightarrow m_1 = \frac{\rho\pi d^2 t}{4}$$

Mass of square plate $m_2 = \rho d^2 t$

$$\therefore m_2 > m_1$$

$\Rightarrow$ COM lies inside the square plate

15. (3)

Sol. M.I. $= MR^2 = m(\sqrt{3/2} b)^2$
 $= m \cdot 3/4 b^2 = 3/4 mb^2$

16. (4)

Sol. As $I = \frac{MR^2}{2} + Mx^2$ or $I \propto x^2$

So, graph between I and x will be a parabola and not passing through origin as for $x = 0$, $I \neq 0$

17. (1)

Sol. Centrifugal force in the string is given by

$$F = \frac{mv^2}{r}$$

$$v^2 = \frac{Fr}{m}$$

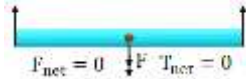
$$v^2 = \frac{25 \times 1.96}{0.25} = 196$$

$$v = \sqrt{196} = 14 \text{ m/s}$$

18. (3)

Sol.

Rod 3 can be in equilibrium in case additional force is applied



19. (1)

Sol. Gravitational force between two masses are independent of medium between them.

20. (3)

Sol. Acceleration due to the gravity $g = \frac{GM}{R^2}$

According to the question, acceleration due to gravity is same on both planets.

$$\therefore \frac{GM_p}{R_p^2} = \frac{GM_e}{R_e^2}$$

$$\Rightarrow \frac{G \times \frac{4}{3} \pi R_p^3 \rho_p}{R_p^2} = \frac{G \times \frac{4}{3} \pi R_e^3 \rho_e}{R_e^2}$$

We get $R_p \rho_p = R_e \rho_e$

Given, $\rho_p = 2\rho_e$

$$\therefore R_p \times 2\rho_e = R_e \rho_e$$

$$\Rightarrow R_p = \frac{R_e}{2} = \frac{R}{2}$$

21. (3)

Torque is the cross product of the force F and the position vector ' r '

$$\vec{\tau} = \vec{r} \times \vec{F}$$

So the angle between τ and F is 90° and between τ and r is 90° .

For example if we take dot product between two vectors as $\theta = 90^\circ$

$$\Rightarrow \cos\theta = \cos 90^\circ = 0 \Rightarrow r \cdot \tau = 0$$

$$F \cdot \tau = 0$$

22. (1)

Sol. P.E = $\frac{GMm}{(R+h)}$

here $h = R$.

Therefore

$$\begin{aligned} \text{P.E} &= \frac{GMm}{(R+R)} = \frac{(GM/2R)m}{1} \\ &= (GM/R^2) \times (mR/2) = \frac{1}{2} mgR \end{aligned}$$

23. (4) 0.80 kgm^2

Sol. $K_r = \frac{1}{2} I \omega^2$

here $K_r = 360 \text{ J}$ and $\omega = 30 \text{ rad/s}$.

24. (1)

Sol. $L_1 = L_2$

$$I_1 \omega_1 = I_2 \omega_2$$

$$\frac{MR^2}{2} \omega_1 \left(\frac{MR^2}{2} + mR^2 + mR^2 \right) \omega_2$$

$$\omega_1 \frac{M}{2} = \frac{(M+4m)}{2} \omega_2$$

$$\omega_2 = \left(\frac{M}{M+4m} \right) \omega_1$$

25. (3)

Sol.

$$x_{CM} = \frac{(m \times 0) + (m \times a) + (2m \times a) + (2m \times 0)}{m + m + 2m + 2m}$$

$$x_{CM} = \frac{ma + 2ma}{6m} = \frac{3ma}{6m} = \frac{a}{2}$$

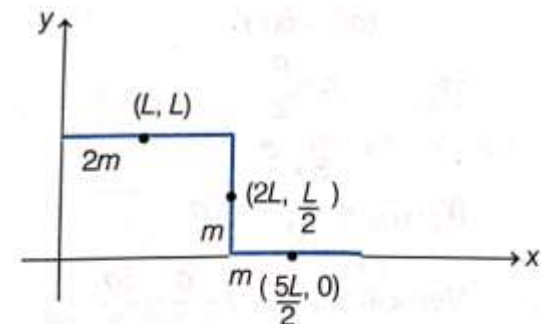
$$y_{CM} = \frac{(m \times 0) + (m \times 0) + (2m \times a) + (2m \times a)}{m + m + 2m + 2m}$$

$$y_{CM} = \frac{4ma}{6m} = \frac{2a}{3}$$

Thus, coordinates of CM is $\left(\frac{a}{2}, \frac{2a}{3} \right)$

26. (1)

Sol.



$$\text{So, } X_{COM} = \frac{2mL + m2L + m \frac{5L}{2}}{4m} = \frac{13}{8} L$$

$$\text{and } Y_{COM} = \frac{2mL + m \times \frac{L}{2} + m \times 0}{4m}$$

$$= \frac{5L}{8}$$

So, position vector of COM is

$$\vec{r}_{COM} = X_{COM}\hat{x} + Y_{COM}\hat{y}$$

$$= \frac{13}{8}L\hat{x} + \frac{5}{8}L\hat{y}$$

27. (4)

Sol. Since, the explosion is due to internal forces, so the motion of CM after explosion will follow the same parabolic path as it would have followed, if there was no explosion.

28. (4)

For translations equilibrium, $\sum \vec{r} = 0$

A – R,

For rotational equilibrium,

$$\sum \vec{r} = 0$$

B – S

Moment of inertia of a body = MK^2

C – P

Torque is required to produce angular accelerations.

D – q

29. (1)

Total initial angular momentum of the two discs is $L_i = I_1\omega_1 + I_2\omega_2$

the moment of inertia I of the system is

$$I = I_1 + I_2$$

Let ω be the final angular speed of the system.

The final angular momentum of the system is

$$L_f = I\omega = (I_1 + I_2)\omega$$

According to law of conservation of angular momentum, we get

$$L_i = L_f$$

$$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega$$

$$\omega = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$$

30. (2)

Initial kinetic energy, $K = \frac{1}{2}I\omega^2$..(i)

According to the principle of conservation of angular momentum, $I\omega = \text{constant}$. As I is doubled, ω becomes half.

So final kinetic energy.

$$K' = \frac{1}{2}(2I)\left(\frac{\omega}{2}\right)^2 = \frac{1}{4}I\omega^2 = \frac{K}{2} \quad (\text{Using (i)})$$

31. (4)

Here, $m = 5 \text{ kg}$, $I = 0.65 \text{ kg m}^2$

$\omega = 10 \text{ rad s}^{-1}$, $R = 0.40 \text{ m}$

Linear velocity

$$v = R\omega = 0.40 \times 10 = 4 \text{ m s}^{-1}$$

Translational KE

$$= \frac{1}{2}mv^2 = \frac{1}{2} \times 5 \times 16 = 40 \text{ J}$$

Rotational KE

$$= \frac{1}{2}I\omega^2 = \frac{1}{2} \times 0.65 \times 100 = 32.5 \text{ J}$$

Total KE = Translational KE + Rotational KE

$$= 40 + 32.5 = 72.5 \text{ J}$$

Percentage of rotational KE

$$= \frac{\text{Rotational KE}}{\text{Total KE}} \times 100$$

$$= \frac{32.5}{72.5} \times 100 = 44.8\%$$

32. (1)

$$I = 4 \left[\frac{M\ell}{12} + \frac{\ell}{2} \right] = 3\ell^2$$

33. (3)

If a body has mass M and radius of gyration is K, then $I = MK^2$

Moment of inertia of a disc and circular ring about a tangential axis in their planes are respectively.

$$I_d = \frac{5}{4}M_d R^2 \Rightarrow I_r = \frac{3}{2}M_r R^2$$

$$\text{but } I = MK^2 \Rightarrow K = \sqrt{\frac{I}{M}}$$

$$\therefore \frac{K_d}{K_r} = \sqrt{\frac{I_d}{I_r} \times \frac{M_r}{M_d}} \quad \text{or}$$

$$\frac{I_d}{I_r} = \sqrt{\frac{(5/4)M_d R^2}{(3/2)M_r R^2} \times \frac{M_r}{M_d}} = \sqrt{\frac{5}{6}}$$

$$\therefore I_d : I_r = \sqrt{5} : \sqrt{6}$$

34. (3)

The direction of L is perpendicular to the line joining the bob to point C. Since this line keeps changing its orientation in space, direction of L keeps changing however as ω is constant, magnitude of L remain constant.

Aliter : The torque about point is perpendicular to the angular momentum vector about point C. Hence it can only change the direction of L , and not its magnitude.

35. (3)

$$\omega = \frac{d\theta}{dt} = \frac{d}{dt}(2t^3 + 0.5) = 6t^2$$

$$\text{at } t = 2 \text{ s, } \omega = 6 \times (2)^2 = 24 \text{ rad/s}$$

36. (2)

Sol. The GPE of an object placed at earth's surface is

$$U_1 = -\frac{GMm}{R} \quad \dots(i)$$

The GPE of object at a height h above the surface of earth is

$$U_2 = -\frac{GMm}{(R+h)} \quad \dots(ii)$$

So, the change in potential energy is

$$\Delta U = U_2 - U_1 = -\frac{GMm}{R+h} - \left(-\frac{GMm}{R}\right)$$

[From Eq. (i) and (ii)]

$$= -GMm \left[\frac{1}{(R+h)} - \frac{1}{R} \right]$$

$$= -GMm \left(-\frac{h}{R(R+h)} \right) = \frac{GMmh}{R(R+h)}$$

37. (1)

Sol. From Kepler's third law, the time period of revolution of satellite around earth is

$$T^2 \propto r^3 \text{ or } T \propto r^{3/2} \dots(i)$$

$$\text{Here, } r_1 = 6R_E + R_E, T_1 = 24 \text{ h}$$

$$r_2 = 2.5R_E + R_E, T_2 = ?$$

$$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2} \right)^{3/2}$$

$$\frac{24}{T_2} = \left(\frac{6R_E + R_E}{2.5R_E + R_E} \right)^{3/2} = \left(\frac{7}{3.5} \right)^{3/2},$$

$$\Rightarrow T_2 = \frac{12}{\sqrt{2}} = 6\sqrt{2} \text{ h}$$

38. (1)

Sol. Since, the escape velocity of earth can be given as

$$v_e = \sqrt{2gR} = R \sqrt{\frac{8}{3} \pi G \rho}$$

[ρ = density of earth]

$$\Rightarrow v_e = R \sqrt{\frac{8}{3} \pi G \rho} \dots \dots (i)$$

As it is given that the radius and mean density of planet are twice as that of earth. So, escape velocity at planet will be

$$v_p = 2R \sqrt{\frac{8}{3} \pi G 2\rho} \dots(ii)$$

Divide, Eq. (i) by Eq. (ii), we get

$$\frac{v_e}{v_p} = \frac{R \sqrt{\frac{8}{3} \pi G \rho}}{2R \sqrt{\frac{8}{3} \pi G 2\rho}}$$

$$\Rightarrow \frac{v_e}{v_p} = \frac{1}{2\sqrt{2}}$$

39. (4)

Sol. The value of acceleration due to gravity is maximum at poles. So, $g_{\text{poles}} > g_{\text{equator}}$

It is always directed towards the center of earth.

40. (2)

$$\text{Sol. } v'_0 = \sqrt{g'r}$$

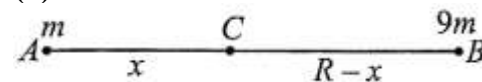
$$\text{and } g' = g \left[1 + \frac{h}{R} \right]^{-2}$$

$$\text{Now, at } h = R/2 \Rightarrow g' = g \left(\frac{2}{3} \right)^2$$

$$v'_0 = \sqrt{g \left(\frac{2}{3} \right)^2 \frac{3R}{2}}$$

$$v'_0 = v_0 \sqrt{\frac{2}{3}}$$

41. (2)



Sol.

Gravitational field at C is zero.

$$\frac{Gm}{x^2} = \frac{G \cdot 9m}{(R-x)^2}$$

$$\frac{R-x}{x} = 3 \Rightarrow x = \frac{R}{4}, R-x = \frac{3R}{4}$$

$$V_g = -G \left[\frac{m}{x} + \frac{9m}{R-x} \right]$$

$$= -Gm \left[\frac{1}{R/4} + \frac{9}{3R/4} \right]$$

$$= -\frac{16Gm}{R}$$

42. (2)

Sol. Due to inertia of direction.

43. (4)

Sol. Kinetic energy = $\frac{GM_E m}{2r}$

A - s

$$\text{Potential energy} = -\frac{GM_E m}{r}$$

B - r

$$\text{Total energy} = -\frac{GM_E m}{2r}$$

C - P

$$\text{Orbital velocity} = \sqrt{\frac{GM_E}{r}}$$

D - q

44. (4)

Sol. As, kinetic energy, $K = \frac{GMm}{2r}$
Potential energy, $U = -\frac{GMm}{r}$

$$\text{Total energy, } E = -\frac{GMm}{2r}$$

If r is decreased, K will increase but U and E will decrease.

45. (3)

Sol. By the principle of superposition of fields

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

Here, $\vec{E}$ = net field at the centre of hole due to entire mass

$\vec{E}_1$ = field due to remaining mass

and $\vec{E}_2$ = field due to mass in hole = 0

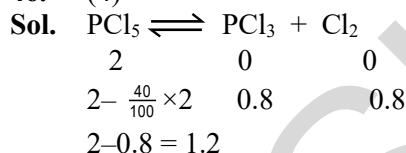
$$\therefore \vec{E}_1 = \vec{E} = \left(\frac{GM}{R^3} \right) \cdot r$$

$$\text{where } r = \frac{R}{2}$$

$$\therefore \vec{E} = \frac{GM}{2R^2}$$

CHEMISTRY

46. (4)



$$\text{So, } [\text{PCl}_5] \frac{1.2}{2} = 0.6$$

$$[\text{PCl}_3] = \frac{0.8}{2} = 0.4$$

$$[\text{Cl}_2] = \frac{0.8}{2} = 0.4$$

$$K_C = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{0.4 \times 0.4}{0.6} = \frac{1.6}{0.6} = 0.267$$

47. (2)

Sol. High press & high temp.

$\text{H}_2\text{O(s)} \rightarrow \text{ice} \rightarrow$ has low density & high volume

$\text{H}_2\text{O(l)} \rightarrow$ has high density & low volume.

So low volume $\rightarrow$ high pressure.

As reaction is endothermic high temperature is required.

48. (1)

Sol. (A - r) ; (B - q) ; (C - p)

(A) Δn_g is +ve so as P is increased, backward shifting will take place.

(B) No change as $\Delta n_g = 0$.

(C) Forward shifting will take place.

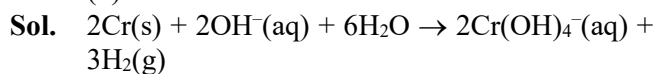
49. (1)

Sol. A catalyst does not influence the values of equilibrium constant but Catalysts influence the rate of both forward and backward reactions equally.

50. (2)

$$\text{Sol. } \frac{a_1}{a_2} = \sqrt{\frac{K_{a_1}}{K_{a_2}}} = \sqrt{\frac{3.14 \times 10^{-4}}{1.96 \times 10^{-5}}} = 4:1$$

51. (2)



$$2 + 2 + 6 + 2 + 3 = 15$$

52. (1)

Sol. x = 2 and acid is monobasic

53. (4)

54. (4)

55. (1)

Sol: $1 \times 10^{-12} = [\text{Mg}^{+2}] [\text{OH}^-]^2$
 $= 0.01 \times [\text{OH}^-]^2$
 $1 \times 10^{-10} = [\text{OH}^-]^2$
 $[\text{OH}^-] = 10^{-5}$
 $\text{pOH} = -\log \text{OH}^- = 5$
 $\text{pH} = 14 - 5 = 9$

56. (2)

Sol: $\text{CH}_3\text{COOH} + \text{HF} \rightleftharpoons \text{CH}_3\text{COOH}_2^+ + \text{F}^-$
 HF gives H^+ to CH_3COOH . So it is a conjugate base of HF.

57. (1)

Sol:

58. (4)

Sol: $2\text{CrCl}_3 + 3\text{H}_2\text{O}_2 + 10\text{NaOH} \rightarrow 2\text{Na}_2\text{CrO}_4 + 6\text{NaCl} + 8\text{H}_2\text{O}$

59. (1)

Sol. Step 1. Write the balanced dissociation equilibrium: $\text{I}_2(\text{g}) \rightleftharpoons 2\text{I}(\text{g})$.
 Step 2. Initial concentrations in a 1.0 L ask:
 $[\text{I}] = 1.0\text{M}$, $[\text{I}_2] = 0$.
 Assume y mol/L of I_2 dissociates at equilibrium.
 Thus equilibrium concentrations are $[\text{I}_2]_{\text{eq}} = 1.0 - y$, $[\text{I}]_{\text{eq}} = 2y$.
 Step 3. Expression for equilibrium constant:

$$K = \frac{[\text{I}]^2}{[\text{I}_2]}$$

Substitute equilibrium values: $K = \frac{(2y)^2}{1.0 - y}$.

Given $K_C = 37.6 \times 10^{-6}$.

Step 4. Because K_C is small, approximate $1.0 - y \approx 1.0$: $4y^2 \approx 37.6 \times 10^{-6}$.

$y^2 \approx 9.4 \times 10^{-6}$, hence $y \approx 3.07 \times 10^{-3}$.

$[\text{I}_2]_{\text{eq}} \approx 1.0 - 0.00307 = 0.99693\text{M}$, $[\text{I}]_{\text{eq}} = 2y \approx 0.00614\text{M}$.

Step 5. Comparison: $[\text{I}_2]_{\text{eq}} = 0.99693\text{M} > [\text{I}]_{\text{eq}} = 0.00614\text{M}$.

Therefore at equilibrium $[\text{I}_2] > [\text{I}]$, so Option 2 A is correct.

60. (1)

Sol. Step 1. Write the balanced reaction and the expression for K_p .

Balanced reaction: $\text{PCl}_{5(\text{g})} \rightleftharpoons \text{PCl}_{3(\text{g})} + \text{Cl}_{2(\text{g})}$.

Expression for K_p : $K_p = \frac{P_{\text{PCl}_3} \times P_{\text{Cl}_2}}{P_{\text{PCl}_5}}$.

Step 2. Calculate the reaction quotient Q_p using initial pressures.

$$Q_p = \frac{(2\text{atm}) \times (2\text{atm})}{2\text{atm}} = 2.$$

Step 3. Compare Q_p and K_p .

Since $Q_p = 2$ and $K_p = 0.512$, $Q_p > K_p$, the equilibrium shifts to the left.

Step 4. Conclude which species increases.

A left shift means more $\text{PCl}_{5(\text{g})}$ is formed. Hence, More PCl_5 will be produced.

61. (1)

Sol. Step 1. On dissolution, CH_3COONa dissociates into CH_3COO^- and Na^+ .

Step 2. The cation Na^+ is neutral, whereas the anion CH_3COO^- undergoes hydrolysis:



Step 3. Define the base hydrolysis constant K_b for CH_3COO^- :

$$K_b = \frac{[\text{CH}_3\text{COOH}][\text{OH}^-]}{[\text{CH}_3\text{COO}^-]}.$$

Step 4. Relate K_b to the acid dissociation constant K_a : $K_b = \frac{K_w}{K_a}$, where K_w is the ionic product of water.

Step 5. For an initial anion concentration c and a small hydrolysis extent x ,

$$[\text{OH}^-] = x, [\text{CH}_3\text{COO}^-] \approx c.$$

$$\text{Then } K_b = \frac{x^2}{c}, \text{ so } x = \sqrt{K_b c} = \sqrt{\frac{K_w c}{K_a}}.$$

$$\text{Step 6. Hence } [\text{OH}^-] = \sqrt{\frac{K_w c}{K_a}}, \text{ pOH} = -\log$$

$$[\text{OH}^-] = \frac{1}{2} (\text{p}K_b - \log c).$$

Step 7. Use the relation $\text{pH} + \text{pOH} = \text{p}K_w$ to get

$$\text{pH} = \text{p}K_w - \text{pOH} = \text{p}K_w - \frac{1}{2} (\text{p}K_b - \log c).$$

Step 8. Since $\text{p}K_a + \text{p}K_b = \text{p}K_w$, substitute to obtain

$$\text{pH} = \frac{1}{2} (\text{p}K_w + \text{p}K_a + \log c)$$

Therefore, the correct expression is $\text{pH} = \frac{1}{2}$
 $(\text{pK}_w + \text{pK}_a + \log c)$, which corresponds to Option A.

62. (1)

Sol. Step 1. Identify the given data: mass of I_2 is 5 g.
 Step 2. Write the balanced redox reaction in acidic medium:



Step 3. Determine the mole ratio: from the equation, 3 moles of I_2 consume 10 moles of HNO_3 ,

so 1 mole of I_2 consumes $\frac{10}{3}$ moles of HNO_3 .

Step 4. Calculate moles of I_2 : $M(\text{I}_2) = 2 \times 126.90 = 253.80 \text{ g/mol}$,

$$n(\text{I}_2) = \frac{5 \text{ g}}{253.80 \text{ g/mol}} \approx 0.01971 \text{ mol}.$$

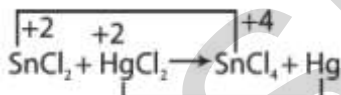
Step 5. Calculate required moles of HNO_3 :

$$n(\text{HNO}_3) = \frac{10}{3} \times 0.01971 \approx 0.06570 \text{ mol}.$$

Step 6. Compute the mass of HNO_3 : $M(\text{HNO}_3) = 14.01 + 1.008 + 3 \times 16.00 = 63.018 \text{ g/mol}$,
 $m(\text{HNO}_3) = 0.06570 \text{ mol} \times 63.018 \text{ g/mol} \approx 4.14 \text{ g}$.

Step 7. Final answer: 4.14 g of HNO_3 (approximately 4.13 g).

63. (4)



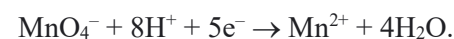
Sol.

64. (2)

Sol. Step 1. Determine the n-factor of KMnO_4 in acidic medium.

Oxidation state of Mn in MnO_4^- is +7 and in Mn^{2+} is +2.

Reduction half-reaction:

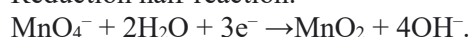


Electrons gained per formula unit = 5, hence $x = 5$.

Step 2. Determine the n-factor of KMnO_4 in strongly basic medium.

Here, MnO_4^- is reduced to MnO_2 where Mn is +4.

Reduction half-reaction:

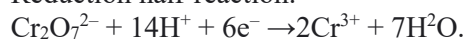


Electrons gained per formula unit = 3, hence $y = 3$.

Step 3. Determine the n-factor of $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium.

Oxidation state of Cr in $\text{Cr}_2\text{O}_7^{2-}$ is +6 and in Cr^{3+} is +3.

Reduction half-reaction:



Electrons gained per formula unit = 6, hence $z = 6$.

Step 4. Calculate $x + y + z$.

Therefore, $x + y + z = 5 + 3 + 6 = 14$.

65. (4)

Sol. Step 1. In the compound $\text{K}_4[\text{Ni}(\text{CN})_4]$, potassium is an alkali metal with oxidation state +1.

Step 2. Each cyanide ligand CN^- has oxidation state -1.

Step 3. Let x be the oxidation state of nickel in the complex.

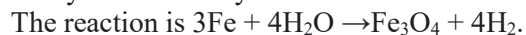
Step 4. For a neutral compound, the sum of oxidation states equals zero: $4(+1) + x + 4(-1) = 0$.

Step 5. Simplify the equation: $4 - 4 + x = 0$, so $x = 0$.

Thus, the oxidation number of nickel in $\text{K}_4[\text{Ni}(\text{CN})_4]$ is 0.

66. (4)

Sol. Step 1: Write the balanced chemical equation to verify stoichiometry.



Step 2: Assign oxidation states of iron in reactants and products.

In the reactant side, each iron atom is in oxidation state 0.

In the product Fe_3O_4 , iron exists in two oxidation states: two are +3 and one is +2.

Step 3: Determine the change in oxidation number for each iron atom.

Each of the two Fe atoms oxidised to Fe^{3+} loses 3 electrons, and the third Fe atom oxidised to Fe^{2+} loses 2 electrons.

Step 4: Calculate the total number of electrons lost.

Total electrons lost = $2 \times 3 + 1 \times 2 = 8$.

Step 5: Conclude the number of electrons transferred. Thus, the total number of electrons lost (and gained by water) in this reaction is 8.

67. (3)

Sol. Step 1. Identify the acid-base character of each species:

Label & Species & Character
A & NH_4Cl & Salt of weak base NH_4OH

B & CH ₃ COONa & Salt of weak acid CH ₃ COOH
C & NH ₄ OH & Weak base
D & CH ₃ COOH & Weak acid

Step 2. Recall buffer requirement: A buffer contains a weak acid with its conjugate base or a weak base with its conjugate acid.

Step 3. Evaluate each given pairing (equal volumes of 1 M solutions yield equal molar amounts):

Option A (C + D): NH₄OH + CH₃COOH – No conjugate acid–base pair; not a buffer.

Option B (A + B): NH₄Cl + CH₃COONa – Two salts without weak partner; not a buffer.

Option C (A + C): NH₄Cl + NH₄OH – Weak base NH₄OH and its conjugate acid NH₄⁺ form a buffer.

Option D (B + C): CH₃COONa + NH₄OH – No conjugate acid–base pair; not a buffer.

Step 4. Conclusion: Only mixing A and C produces a buffer solution.

68. (4)

Sol. Step 1. Definition of pH. pH is defined as $\text{pH} = -\log[\text{H}^+]$, where $[\text{H}^+]$ denotes the hydrogen ion concentration in mol L^{-1} .

Step 2. Individual concentrations.

(i) For pH 3, $[\text{H}^+] = 10^{-3} \text{ M}$.

(ii) For pH 4, $[\text{H}^+] = 10^{-4} \text{ M}$.

(iii) For pH 5, $[\text{H}^+] = 10^{-5} \text{ M}$.

Step 3. Total moles of H^+ when equal volumes are mixed.

Let each volume be V_0 . Moles contributed = $V_0 (10^{-3} + 10^{-4} + 10^{-5})$.

Step 4. Final concentration after mixing.

$$\text{Total volume} = 3V_0. \text{ Thus, } [\text{H}^+]_{\text{mixture}} = \frac{V_0 (10^{-3} + 10^{-4} + 10^{-5})}{3V_0} = \frac{10^{-3} + 10^{-4} + 10^{-5}}{3} \dots$$

Step 5. Numerical calculation.

Sum in numerator = $10^{-3} + 10^{-4} + 10^{-5} =$

$$0.00111. \text{ Dividing by 3 gives } \frac{0.00111}{3} =$$

0.00037. 3

Step 6. Result. Therefore, $[\text{H}^+]_{\text{mixture}} = 3.7 \times 10^{-4} \text{ M}$.

69. (1)

Sol. Step One: List the given data.

Mass of $\text{H}_2\text{AO}_4 = 0.135 \text{ g}$, and equivalents of NaOH required = $2.75 \times 10^{-3} \text{ eq}$.

Step Two: Calculate the equivalent mass of the acid.

$$\text{Equivalent mass} = \frac{0.135 \text{ g}}{2.75 \times 10^{-3} \text{ eq}} \approx 49.09 \text{ g/eq}.$$

Step Three: Determine the molar mass using acid basicity.

Since the acid is diprotic, molar mass = (equivalent mass) $\times 2 = 49.09 \times 2 \approx 98.18 \text{ g/mol}$.

Step Four: Express the molar mass in terms of $M(\text{A})$.

$$M(\text{H}_2\text{AO}_4) = 2 \times 1.008 + M(\text{A}) + 4 \times 16.00 = M(\text{A}) + 66.016.$$

Step Five: Solve for $M(\text{A})$.

$$M(\text{A}) + 66.016 = 98.18 \Rightarrow M(\text{A}) = 98.18 - 66.016 = 32.164 \text{ u} \approx 32 \text{ u}.$$

Step Six: Relate atomic mass to neutrons and protons.

If the number of neutrons and protons are equal, mass number = $2Z \approx 32 \Rightarrow Z = 16$.

Therefore, the number of neutrons = 16.

70. (2)

Sol. Step 1. The compounds KH, MgH_2 and NaH are ionic 2 metal hydrides.

Step 2. In these hydrides, hydrogen is more electronegative than the metal and exists as hydride ion with oxidation number -1.

Step 3. Applying the oxidation number sum rule: Compound & Metal oxidation number & Hydrogen oxidation number

KH & +1 & -1

MgH_2 & +2 & -1

NaH & +1 & -1

Step 4. Hence, the oxidation number of hydrogen in each compound is -1.

Step 5. The correct order is -1, -1, -1, corresponding to option B.

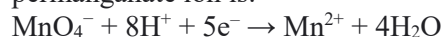
71. (1)

Sol. Fe_3O_4 is a mixed oxide of $\text{FeO} \cdot \text{Fe}_2\text{O}_3$.

72. (3)

Sol. Step 1. Equivalent weight definition: It is the molar mass divided by the number of electrons transferred per molecule in a redox reaction.

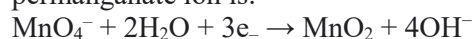
Step 2. In acidic medium, the reduction of permanganate ion is:



Here, each MnO_4^- gains 5 electrons, so $n = 5$.

$$\text{Therefore, equivalent weight} = \frac{M}{5}.$$

Step 3. In neutral medium, the reduction of permanganate ion is:



Here, each MnO_4^- gains 3 electrons, so $n = 4$.

Therefore, equivalent weight = $\frac{M}{3}$.

Step 4. Comparing with $\frac{M}{a}$ and $\frac{M}{b}$, we find a

= a = 5, b = 3.

Hence, the values of a and b are 5 and 3 respectively, which corresponds to Option C.

73. (2)

Sol. Step 1. Determine oxidation state of M in the product:

In MO_3^- , oxygen is -2, so total O charge = $3 \times (-2) = -6$;

to give overall charge -1, M must be +5.

Thus $\text{M}^{x+} \rightarrow \text{M}^{5+}$, electrons lost per M = $5 - x$.

Step 2. Determine electrons gained by permanganate:

Reduction of $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ gains 5 electrons per ion.

Step 3. For 2 moles of MnO_4^- : total electrons gained = $2 \times 5 = 10$.

Step 4. Moles of M oxidized by 2 permanganate = $2 \times 2.5 = 5$.

Electrons lost by 5 moles of M = $5(5 - x)$ must equal 10.

$5(5 - x) = 10 \Rightarrow 25 - 5x = 10 \Rightarrow 5x = 15 \Rightarrow x = 3$.

74. (3)

Sol. Step One: Identify the relevant equation. The Henderson-Hasselbalch equation for an acid buffer is

$$\text{pH} = \text{pK}_a + \log \frac{[\text{HCOO}^-]}{[\text{HCOOH}]}$$

Here $[\text{HCOO}^-]$ comes from HCOONa and $[\text{HCOOH}]$ is formic acid.

Step Two: Compute pK_a for formic acid.

$$\text{K}_a(\text{HCOOH}) = 1.8 \times 10^{-4}$$

$$\Rightarrow \text{pK}_a = -\log(1.8 \times 10^{-4}) \approx 3.745$$

Step Three: Substitute pH and pK_a into the equation.

$$4.25 = 3.745 + \log \frac{[\text{HCOO}^-]}{[\text{HCOOH}]}$$

$$\log \frac{[\text{HCOO}^-]}{[\text{HCOOH}]} = 4.25 - 3.745 = 0.505$$

Step Four: Find the ratio by taking antilogarithm.

$$\frac{[\text{HCOO}^-]}{[\text{HCOOH}]} = 10^{0.505} \approx 3.24$$

Therefore, the ratio of HCOONa to HCOOH required for pH 4.25 is 3.24.

75. (3)

Sol. Given reaction: $3\text{ClO}^- \rightarrow \text{ClO}_3^- + 2\text{Cl}^-$.

Step 1. Determine oxidation states.

In ClO^- , let oxidation state of Cl be x. Then $x + (-2) = -1$ gives $x = +1$.

In ClO_3^- , let oxidation state of Cl be y. Then $y + 3 \times (-2) = -1$ gives $y = +5$.

In Cl^- , oxidation state of Cl is -1.

Step 2. Identify oxidation and reduction.

Change from +1 to +5 represents oxidation (increase in oxidation state).

Change from +1 to -1 represents reduction (decrease in oxidation state).

Step 3. Classify the reaction.

76. (2)

Sol. Step 1. Solubility product for each sulphide.

(a) For CuS : CuS(s) dissociates into Cu^{2+} and S^{2-} ,
 $\text{K}_{\text{sp}} = [\text{Cu}^{2+}][\text{S}^{2-}] = s \times s = s^2$.

Step 2. Solubility of CuS .

$$\text{K}_{\text{sp}}(\text{CuS}) = 10^{-31} \Rightarrow s(\text{CuS}) = \sqrt{10^{-31}} = 10^{-15.5}$$

(b) For Ag_2S : $\text{Ag}_2\text{S(s)}$ dissociates into 2Ag^+ and S^{2-} ,

$$\text{K}_{\text{sp}} = [\text{Ag}^+]^2[\text{S}^{2-}] = (2s)^2 \times s = 4s^3$$

Step 3. Solubility of Ag_2S .

$$\text{K}_{\text{sp}}(\text{Ag}_2\text{S}) = 10^{-44} \Rightarrow s(\text{Ag}_2\text{S}) = \left(\frac{10^{-44}}{4} \right)^{1/3}$$

$$\approx 1.36 \times 10^{-15}$$

(c) For HgS : HgS(s) dissociates into Hg^{2+} and S^{2-} ,
 $\text{K}_{\text{sp}} = [\text{Hg}^{2+}][\text{S}^{2-}] = s^2$.

Step 4. Solubility of HgS .

$$\text{K}_{\text{sp}}(\text{HgS}) = 10^{-54} \Rightarrow s(\text{HgS}) = \sqrt{10^{-54}} = 10^{-27}$$

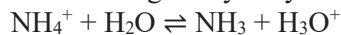
Step 5. Comparison of solubilities.

$$\text{HgS} : 10^{-27} < \text{CuS} : 10^{-15.5} < \text{Ag}_2\text{S} : 1.36 \times 10^{-15}$$

Therefore, the solubilities increase in the order $\text{HgS} < \text{CuS} < \text{Ag}_2\text{S}$.

77. (4)

Sol. Step 1: In aqueous solution, the ammonium ion NH_4^+ undergoes hydrolysis as follows: 4



Step 2: The hydrolysis constant K_h for this reaction is related to the ionization constant K_b of NH_4OH by:

$$\text{K}_h = \frac{\text{K}_w}{\text{K}_b}$$

Step 3: At 298 K, $\text{K}_w = 1.0 \times 10^{-14}$ and $\text{K}_b = 1.77 \times 10^{-5}$.

Step 4: Substituting these values gives:

$$K_h = \frac{1.0 \times 10^{-14}}{1.77 \times 10^{-5}}$$

Step 5: Evaluating the numerical and exponent

parts, $\frac{1.0}{1.77} \approx 0.565$, and $\frac{10^{-14}}{10^{-5}} = 10^{-9}$, hence:

$$K_h \approx 0.565 \times 10^{-9} = 5.65 \times 10^{-10}$$

Step 6: Therefore, the hydrolysis constant of ammonium chloride is 5.65×10^{-10} .

78. (1)

Sol. Step 1. The van't Hoff equation relates the equilibrium constant K to temperature:

$$\ln K = -\frac{\Delta H^\circ}{R} \frac{1}{T} + \frac{\Delta S^\circ}{R}$$

Step 2. For an exothermic reaction, $\Delta H^\circ < 0$.

Step 3. Hence the slope of the plot $\ln K$ vs $\frac{1}{T}$ is

$$-\frac{\Delta H^\circ}{R}, \text{ which is positive.}$$

Step 4. On the given graph, the lines with positive slopes correspond to exothermic behaviour.

Step 5. Lines A (vertical line indicating a very large positive slope) and B (moderate positive slope) have positive slopes.

79. (4)

Sol. In case of HCl, $M = N$ (n.factor = 1)
 millimoles of H^+ from 20ml, 0.1N HCl = 2 millimoles
 millimoles of H^+ from 500ml, 0.1 M HCl (pH = 1) = 50 millimoles
 $[H^+]$ of mixture = $(M_1V_1 + M_2V_2) / (V_1 + V_2)$
 $= (20 \times 0.1 + 500 \times 0.1) / 20 + 500$
 $= 50.2 / 520$
 $= 0.1$
 So, pH = 1

80. (3)

Sol. Step 1. Identify the given data : Moles of $CH_3NH_2 = 0.12$, $pK_b = 3.3$, moles of HCl = 0.08, total volume = 1 L.
 Step 2. Write the neutralisation reaction:
 $CH_3NH_2 + HCl \rightarrow CH_3NH_3^+ + Cl^-$
 Moles of $CH_3NH_3^+$ formed = 0.08, remaining moles of $CH_3NH_2 = 0.12 - 0.08 = 0.04$.
 Step 3. Calculate molar concentrations (in 1 L) :
 $[CH_3NH_2] = 0.04$ M, $[CH_3NH_3^+] = 0.08$ M.
 Step 4. Apply Henderson-Hasselbalch equation for a basic buffer:

$$\begin{aligned} pH &= 14 - pK_b + \log \frac{[CH_3NH_2]}{[CH_3NH_3^+]} \\ &= 14 - 3.3 + \log \frac{0.04}{0.08} = 10.7 + \log 0.5 \\ &= 10.7 - 0.3010 = 10.399 \approx 10.40. \\ \text{Hence, the pH of the solution is 10.4.} \end{aligned}$$

81. (4)

Sol. Step 1. A Bronsted acid is a proton donor and a Bronsted base is a proton acceptor.

Step 2. An amphiprotic species can act both as a proton donor and a proton acceptor.

Step 3. Analysis of each option:

(i) HSO_4^- : can donate a proton to form SO and can accept a proton to form H_2SO_4 , hence amphiprotic.

(ii) HCO_3^- : can donate a proton to form CO_3^{2-} and can accept a proton to form H_2CO_3 , hence amphiprotic.

(iii) NH_3 : can accept a proton to form NH_4^+ but does not donate a proton in aqueous solution, so it acts only as a Bronsted base.

(iv) HCl: can donate a proton to form Cl^- but cannot accept a proton, so it acts only as a Bronsted acid.

Step 4. Conclusion: The species that cannot act both as Bronsted acid and base is HCl.

82. (4)

Sol. Step 1. Definition: A redox reaction involves change in oxidation states of atoms due to electron transfer.

Step 2. Procedure: Assign oxidation numbers to relevant atoms in reactants and products.

Step 3. Option Analysis:

Option A : In XeF_2 , F is -1 so Xe is $+2$; in $[XeF]^+$, Xe remains $+2$. In PF_5 and PF_6^- , P remains $+5$. No change in oxidation states, hence not redox.

Option B: In XeF_6 , Xe is $+6$; in $XeOF_4$, O is -2 and each F is -1 , total -6 , so Xe is $+6$.

All oxidation states unchanged, not redox.

Option C: In XeF_6 , Xe is $+6$; in XeO_2F_4 , each O is -2 and each F is -1 , total -8 , so Xe appears $+8$. Xe oxidation changes without complementary reduction, overall not a redox reaction.

Option D: In XeF_4 , Xe is $+4$; in XeF_6 , Xe is $+6$, so Xe is oxidized. In O_2F_2 , each O is $+1$, while in O_2 it is 0 , so O is reduced. Both oxidation and reduction occur, hence redox.

Step 4. Conclusion: Option D is the only redox reaction.

83. (1)

Sol. Step 1. Write the oxidation half-reaction:

Step 2. Write the reduction half-reaction for KMnO_4 in acidic medium: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$.

Step 3. Determine moles of Fe^{2+} oxidized per mole of KMnO_4 : Each MnO_4^- consumes 5 electrons, so 1 mol KMnO_4 oxidizes 5 mol Fe^{2+} .

Step 4. Calculate volume of 1 M KMnO_4 needed for 1 mol Fe^{2+} :

$$V_{\text{KMnO}_4} = \frac{1 \text{ mol Fe}^{2+}}{5 \left(\frac{\text{mol Fe}^{2+}}{\text{mol KMnO}_4} \right)} \times \frac{1 \text{ L}}{1 \text{ mol KMnO}_4} = 0.20 \text{ L}$$

Step 5. Write the reduction half-reaction for $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$.

Step 6. Determine moles of Fe^{2+} oxidized per mole of $\text{K}_2\text{Cr}_2\text{O}_7$: Each $\text{Cr}_2\text{O}_7^{2-}$ consumes 6 electrons, so 1 mol $\text{K}_2\text{Cr}_2\text{O}_7$ oxidizes 6 mol Fe^{2+} .

Step 7. Calculate volume of 1 M $\text{K}_2\text{Cr}_2\text{O}_7$ needed for 1 mol Fe^{2+} :

Volume

$$V_{\text{K}_2\text{Cr}_2\text{O}_7} = \frac{1 \text{ mol Fe}^{2+}}{6 \left(\frac{\text{mol Fe}^{2+}}{\text{mol K}_2\text{Cr}_2\text{O}_7} \right)} \times \frac{1 \text{ L}}{1 \text{ mol K}_2\text{Cr}_2\text{O}_7} = 0.1667 \text{ L}$$

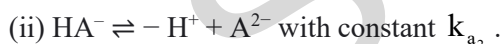
Step 8. Compare the volumes: $0.20 \text{ L} > 0.1667 \text{ L}$.

Therefore, $V_{\text{KMnO}_4} > V_{\text{K}_2\text{Cr}_2\text{O}_7}$

84. (2)

Sol. Step 1: A dibasic acid is an acid that can release two protons sequentially. Consider a generic dibasic acid H_2A .

Step 2: The two dissociation equilibria and their constants are:



Step 3: Upon removal of the first proton, the intermediate ion HA^- carries a single negative charge, whereas removal of the second proton yields A^{2-} with a higher negative charge.

Higher negative charge increases electrostatic attraction between the base and the proton, making the second deprotonation harder.

Hence, $k_{a_1} > k_{a_2}$.

Step 4: Evaluation of the options:

Option A: $k_{a_2} = k_{a_1}$ is incorrect because the first deprotonation is always stronger than the second in a dibasic acid.

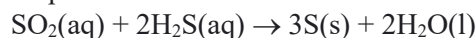
Option B: $k_{a_1} > k_{a_2}$ is correct as shown above.

Option C: $k_{a_1} < k_{a_2}$ is incorrect as it contradicts the electrostatic argument.

Option D: $k_{a_1} = \frac{1}{k_{a_2}}$ is incorrect since there is

no inverse relation between the two dissociation constants.

85. (3)

Sol. Step 1. Write the balanced chemical equation:

Step 2. Assign oxidation states of sulfur in each species:

In SO_2 , let the oxidation state of S be x : $x + 2(-2) = 0 \Rightarrow x = +4$.

In H_2S , let the oxidation state of S be y : $y + 2(+1) = 0 \Rightarrow y = -2$.

In elemental sulfur $\text{S}(\text{s})$, the oxidation state of S is 0.

Step 3. Identify oxidations and reductions:

H_2S : $-2 \rightarrow 0$ (increase in oxidation state = oxidation)

SO_2 : $+4 \rightarrow 0$ (decrease in oxidation state = reduction)

Step 4. Determine the roles:

H_2S loses electrons and thus acts as a reducing agent.

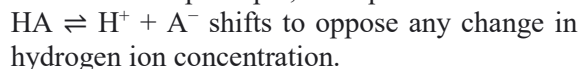
SO_2 gains electrons and thus acts as an oxidizing agent.

Step 5. Conclusion: SO_2 acts as the oxidizing agent in this reaction.

86. (1)

Sol. Step 1. Definition of a buffer solution: a mixture of a weak acid and its conjugate base, or a weak base and its conjugate acid, that resists changes in pH upon addition of small amounts of acid or base.

Step 2. Principle of buffer action: according to Le Chatelier's principle, the equilibrium



When an acid is added (increase in H^+), the conjugate base reacts: $\text{A}^- + \text{H}^+ \rightarrow \text{HA}$.

When a base is added (increase in OH^-), the weak acid reacts: $\text{H}_\text{A} + \text{OH}^- \rightarrow \text{A}^- + \text{H}_2\text{O}$.

These reactions consume the added H^+ or OH^- ions, keeping the $[\text{H}^+]$ nearly constant.

Step 3. Evaluation of each option:

Option A: Correct. The buffer components convert added acid or base into their unionized form, thereby resisting pH change.

Option B: Incorrect. Buffer action arises from chemical equilibrium shifts, not from any shielding by other ions.

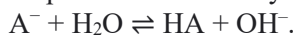
Option C: Incorrect. Buffers do not contain excess free H^+ or OH^- ; they contain a weak acid or base together with its salt.

Option D: Incorrect because only Option A accurately describes the fundamental mechanism of buffer action.

Step 4. Conclusion: The correct answer is Option A.

87. (1)

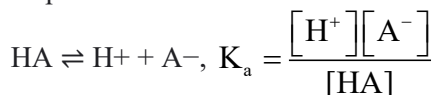
Sol. Step 1. Consider the hydrolysis reaction:



Step 2. Define the hydrolysis constant K_h :

$$K_h = \frac{[HA][OH^-]}{[A^-]}$$

Step 3. Write the acid dissociation constant for HA:



Step 4. Recall the ionic product of water K_w :

$$K_w = [H^+][OH^-]$$

Step 5. Multiply K_h and K_a :

$$K_h K_a = \frac{[HA][OH^-]}{[A^-]} \times \frac{[H^+][A^-]}{[HA]} = K_w$$

Step 6. Solve for K_h :

$$K_h = \frac{K_w}{K_a}$$

Therefore, the hydrolysis constant is $\frac{K_w}{K_a}$,

matching Option A.

88. (3)

Sol. Step 1. Oxidation state concept: the sum of oxidation numbers equals the overall charge of the species.

Step 2. Assign oxidation numbers to nitrogen:

(i) For NH_3 : Let the oxidation state of N be x. H is +1, so $x + 3(+1) = 0$, hence $x = -3$.

(ii) For HNO_3 : Let N be x. H is +1, O is -2, so $+1 + x + 3(-2) = 0$, giving $x = +5$.

(iii) For NaN_3 : Na is +1 and the azide ion is N_3^- , so the sum of oxidation numbers for N is -1. Resonance structures assign two N at -1 and one N at +1, therefore negative oxidation states occur.

(iv) For Mg_3N_2 : Mg is +2, so $3(+2) + 2x = 0$ gives $2x = -6$, thus $x = -3$.

Step 3. Species with negative N oxidation state are NH_3 , NaN_3 , Mg_3N_2 .

Step 4. Therefore the number of such molecules is 3.

89. (1)

Sol. Step 1. Identify the equilibrium:



Step 2. Applying Le Chatelier's principle: an increase in pressure favours the side with lower volume.

Step 3. Densities: ice has density 0.917 g/cm^3 and liquid water has density 1.000 g/cm^3 .

Step 4. Molar mass of H_2O is 18.015 g/mol .

Step 5. Molar volumes:

$$V_m(s) = \frac{18.015 \text{ g/mol}}{0.917 \text{ g/cm}^3}$$

$$\approx 19.65 \text{ cm}^3/\text{mol}, V_m(l) = \frac{18.015 \text{ g/mol}}{1.000 \text{ g/cm}^3} =$$

$$18.015 \text{ cm}^3/\text{mol}$$

Step 6. Since $V_m(l) < V_m(s)$, increased pressure shifts equilibrium towards liquid H_2O .

Step 7. Conclusion: More liquid water will form under increased pressure.

90. (2)

Sol. Step One. Identify the given equilibrium reaction and enthalpy change.

The reaction is $2H_2O(g) \leftrightarrow 2H_2(g) + O_2(g)$ with $\Delta H = +241.7 \text{ kJ}$, indicating the forward reaction is endothermic. Step Two. Effect of temperature according to Le Chatelier's principle.

Increasing temperature shifts equilibrium towards the endothermic forward reaction, increasing decomposition of $H_2O(g)$.

Step Three. Effect of pressure according to Le Chatelier's principle.

Decreasing pressure shifts equilibrium towards the side with more gas moles (forward reaction), increasing decomposition.

Step Four. Option A: Increasing temperature and pressure.

Increasing temperature favours the forward endothermic reaction: $2H_2O(g) \rightarrow 2H_{2(g)} + O_{2(g)}$. Increasing pressure shifts equilibrium towards fewer gas molecules (backward reaction), so these effects oppose each other and maximum decomposition does not occur.

Step Five. Option B: Increasing temperature and decreasing pressure. Both increasing temperature and decreasing pressure favour the forward endothermic reaction and the side with more gas moles, acting synergistically to maximise decomposition of $H_2O(g)$. Step Six.

Option C: Decreasing both temperature and pressure.

Decreasing temperature favours the exothermic backward reaction while decreasing pressure favours the forward reaction; these conflicting effects do not produce maximum decomposition.

Step Seven. Option D: Keeping temperature and pressure constant. No change in conditions results in no shift from the initial equilibrium, so decomposition is unchanged.

Step Eight. Conclusion.

Maximum decomposition of $\text{H}_2\text{O}(\text{g})$ occurs when temperature is increased and pressure is decreased (Option B).

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